

Chapter 4

Petrophysical NMR measurements

4.1 NMR in porous media

As described in earlier chapters, the nuclear magnetic relaxation behavior of water in porous media tends to be quite different from that of water in the bulk phase. There were experiments [41, 37] performed which demonstrated that water, when it resides in the pore space of a rock (e.g. Berea sandstone), has a much shorter relaxation time (T_1 typically varies from 10 to several hundred milliseconds) than it has in the bulk phase (about 3 seconds). The very same water, when removed from the rock, regains its long relaxation time. Hence, the water that came into contact with the Berea sandstone did not change, or at least did not change significantly enough to have any discernible change in the relaxation time. Accordingly, one can only interpret the fast relaxation of water in Berea sandstones as the result of enhanced surface relaxation at the pore-matrix interface, which is presumably due to a combination of paramagnetic impurity ions and reduced correlation times at the interface.

In porous rocks, we mostly study the relaxation of hydrogen nuclei in the pore space. The values of T_1 and T_2 are affected by the physical and chemical properties of the rock fluids, the uniformity of the static magnetic field, the variations in pore sizes, the differences between the magnetic susceptibility of pore fluids and rock grains, the fluid viscosity and diffusion, and the presence of paramagnetic impurity ions on the pore walls or in the fluids.

4.1.1 Surface relaxation

The basic physics of NMR relaxation of fluids at solid surfaces in porous media were first studied by Korringa, SeEVERS, and Torrey [129]. They identified two surface relaxation mechanisms: one occurs at all sites on the surface, and the other is associated with dilute paramagnetic ion impurities. The relaxation process associated with all non-magnetic sites on the surface is much too weak to account for the fast relaxation observed for water in rocks (Almagor and Belfort [6], D'Orazio et al. [73]). Para- and ferromagnetic ions, such as iron, manganese, nickel, and chromium can cause strong surface relaxation. Their presence on the solid surfaces of the rock will dominate the relaxation process of water in the pore space. Typical sandstones have about 1% iron content, and some contain a significant concentration of manganese. This level of paramagnetic impurity ions will make spins relax very efficiently. Carbonates, on the other hand, have more varied mineralogies, large vuggy pores, and a wider range of surface relaxation strength (typically smaller than sandstones). Hence, their relaxation times can sometimes be as long as the bulk water values.

Fast, intermediate, and slow diffusion regimes

There is evidence that the relaxation of polarized proton spins in fluids, which occupy porous rocks or biological tissues, cannot be described by a single relaxation time. Brownstein and Tarr [45] demonstrated that multiexponential decay arises simply as a consequence of an eigenvalue problem of spin diffusion in porous media of various sizes and shapes, even in the case of a very simple shape of an isolated void or biological cell. They considered two properties which govern the evolution of spin polarization in the system: (1) paramagnetic materials on the solid/fluid interface acting as relaxation centers, and (2) classical diffusion of water molecules with the diffusion constant characteristic of bulk water.

The governing equations for diffusion of spins in the porous media can then be written as

$$\begin{aligned} D\nabla^2 m(\mathbf{r}, t) - \frac{m(\mathbf{r}, t)}{T_B} &= \frac{\partial m(\mathbf{r}, t)}{\partial t} \quad \text{in the pore space} \\ \hat{\mathbf{n}} \cdot D\nabla m(\mathbf{r}, t) + \rho m(\mathbf{r}, t) \Big|_S &= 0 \quad \text{at the interface} \end{aligned} \quad (4.1)$$

where $m(\mathbf{r}, t)$ is the density of unrelaxed spins, $1/T_B$ is the bulk relaxation

rate, ρ is the relaxation rate at the solid/fluid interface, or surface relaxation strength, D is the diffusion coefficient of spins in the pore fluid, and $\hat{\mathbf{n}}$ is the unit outward normal on that interface (i.e., it points out of the pore space and into the solid matrix). Brownstein and Tarr showed that the magnetization

$$M(t) = \int m(\mathbf{r}, t) d^3\mathbf{r} \quad (4.2)$$

is given by a sum of decreasing exponential functions of time

$$M(t) = M(0) \sum_{n=0}^{\infty} f_n e^{-\lambda_n t} = M(0) \sum_{n=0}^{\infty} f_n e^{-t/T_n} \quad (4.3)$$

where f_n are the relative amplitudes for each exponential decay, and λ_n (or $1/T_n$) are the eigenvalues of the time independent version of Eq. (4.1), i.e.,

$$D\nabla^2\psi_n(\mathbf{r}) + \lambda_n\psi_n(\mathbf{r}) = 0 \quad (4.4)$$

with the boundary condition

$$\hat{\mathbf{n}} \cdot D\nabla\psi_n(\mathbf{r}) + \rho\psi_n(\mathbf{r}) \Big|_S = 0 \quad (4.5)$$

where the bulk relaxation e^{-t/T_B} has been factored out. (Note that the bulk relaxation term is often neglected in the discussion because typically $T_B \gg T_n$ and the relaxation is dominated by $1/T_n$. However, this is not always true as we shall discuss below.) These eigenvalues, λ_n ($1/T_n$), are primarily determined by the geometry of the pore space. However, their values depend strongly on ρ , the surface relaxivity, and D , the diffusion constant of water.

Brownstein and Tarr found that when $\rho a/D \ll 1$, where a is a characteristic dimension of the system they considered, the lowest eigenvalue, $1/T_0$, is given by

$$\frac{1}{T_0} \cong \rho \frac{S}{V} \equiv \frac{\rho}{a} \quad (4.6)$$

where S is the pore surface and V is the pore volume. a can now be alternatively defined as V/S , a characteristic dimension of the porous system. Thus, the condition $\rho a/D \ll 1$ occurs when the time it takes for the spin to diffuse to the solid/fluid interface, $t \sim a^2/D$, is much shorter than the spin relaxation time, a/ρ , i.e., a fast diffusion situation.

Based upon the three simple pore shapes which they studied, namely parallel slab, circular cylinder, and sphere, Brownstein and Tarr categorized the diffusion into three different regimes, i.e.,

$$\begin{array}{ll} \text{fast-} & \rho a/D \ll 1, \\ \text{intermediate-} & 1 \ll \rho a/D \ll 10, \\ \text{and} & \text{slow-} \quad 10 \ll \rho a/D \end{array}$$

diffusion regimes.

In the *fast* diffusion regime, there is a large gap between the lowest eigenvalue, $\rho S/V$, and the second lowest and higher eigenvalues. Thus, the lowest eigenvalue is the dominant term, and it depends only on S/V : No other detail of the microstructure enters. This is because the fast diffusion equalizes the spin density everywhere within the pore space. The fine details of the pore geometry thus do not play a role in the magnetization decay, except at extremely short times.

In the *slow* diffusion regime, the lowest eigenvalue is of the order of a^2/D , and expectedly, somewhat different for different pore shapes. Its value is very close to the higher eigenvalues which contain detailed information on the pore geometry. In this case, the magnetization decay has a multiexponential behavior, even for a single pore.

Figure 4.1 illustrates the fast and slow diffusion behavior in a simple pictorial way. On the left is the *fast diffusion regime*. Because of the fast diffusion and the small surface relaxivity, the unrelaxed spin density is uniform over the pore space. When these spins wander around and come into contact with the surface and relax, the unrelaxed spin population is quickly equalized through the diffusion process. As a result, the magnetization decays with a uniform spin density. On the right is the *slow diffusion regime*. In this case, the surface relaxivity is large and the diffusion is slow. The unrelaxed spin density near the surface decreases faster than the rate at which the unrelaxed spins can flow in, by diffusion from other parts of the pore space, to replace the loss. The spin density thus develops a profile which is high at the center of the pore and near zero at the pore surface.

The temperature and pressure effects

If rocks are in the slow diffusion regime, then their relaxation rates should depend on the self diffusion coefficient of water which is very sensitive to

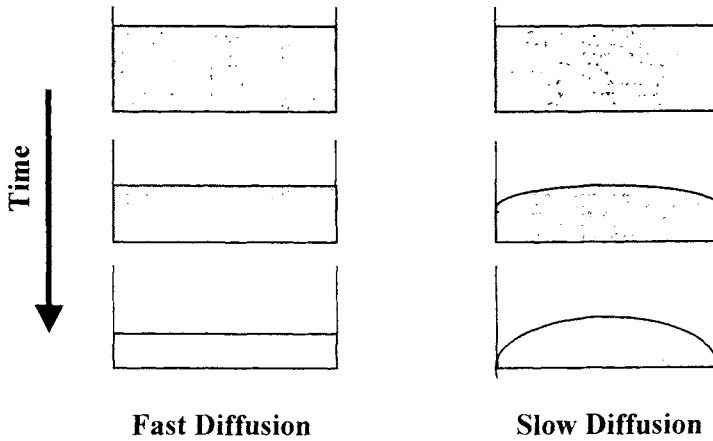


Figure 4.1: Schematic diagram of the slow and fast diffusion regimes

temperature changes. Latour, Kleinberg, and Sezginer [138] have demonstrated experimentally that T_1 and T_2 of water-saturated sandstones and some carbonates are only weakly dependent on temperature over the range of 25° to 175°C. Figure 4.2 shows some of their results. The lack of temperature dependence of relaxation times has been taken as evidence that water-saturated rocks are in the *fast diffusion regime*. It also suggests that the surface relaxivity is independent of temperature.

In the same paper, Latour et al. [138] also demonstrated that the T_1 relaxation behavior of oil-saturated rocks cannot be properly explained by the slow diffusion analysis. If the fast diffusion regime is assumed, they found $\rho_{oil} = \rho_w/17$ in one case, and $\rho_{oil} = \rho_w/30$ in another. However, there is a significant difference between the temperature behavior of the T_1 distributions of water-saturated rocks versus that of oil-saturated rocks. For water-saturated rocks, the surface relaxation plays a dominant role, and the bulk relaxation term can practically be neglected. Hence, if the surface relaxivity is temperature independent, so is the T_1 distribution. This is not true in the oil-saturated rocks. Since $\rho_{oil} \ll \rho_w$, the surface relaxation is very weak, and the oil bulk relaxation, which is temperature dependent, cannot be neglected and has to be included in the consideration. As a result, even

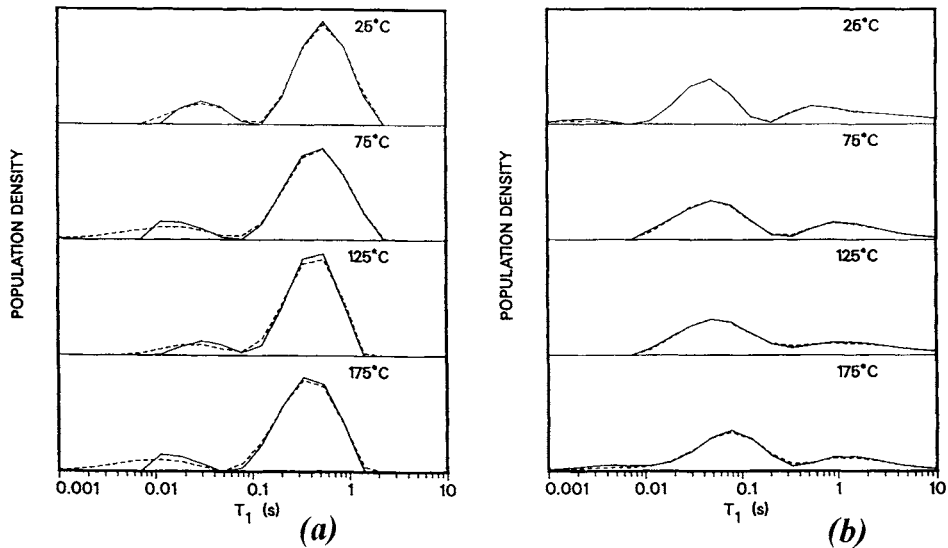


Figure 4.2: The T_1 distributions of water-saturated sandstones (a) and carbonates (b) at four different temperatures. (From Latour et al. [138].)

though the surface relaxation is in the fast diffusion regime and independent of temperature, the overall T_1 distributions, when including the temperature sensitive bulk relaxation, can be quite sensitive to temperature changes.

Recent work on carbonates by Ramakrishnan et al. [183, 184] and Fordham et al. [86] indicated that the carbonates which Latour et al. [138] studied are only a subset of carbonates whose T_1 and T_2 distributions are weakly temperature dependent. They found that although that the T_2 distributions of most sandstones and some carbonates that Latour et al. studied are weakly temperature dependent, there are other types of carbonates whose T_2 distributions show strong temperature dependency. Figure 4.3 shows one such type of carbonate sample which displays a strong temperature dependence in its T_2 distributions. They suggested a model which explains the temperature dependent behavior of the T_2 distributions of some types of carbonates, but fails to explain the others (see discussion on carbonates in Section 4.1.3).

Chen et al. [57] studied the hydrostatic pressure effect on proton and deuteron magnetic relaxation behavior of water in rocks and glass bead packs. From ambient to 250 MPa, they found very little change in the T_1 distributions of H_2O and D_2O saturated porous systems. The weak pressure depen-

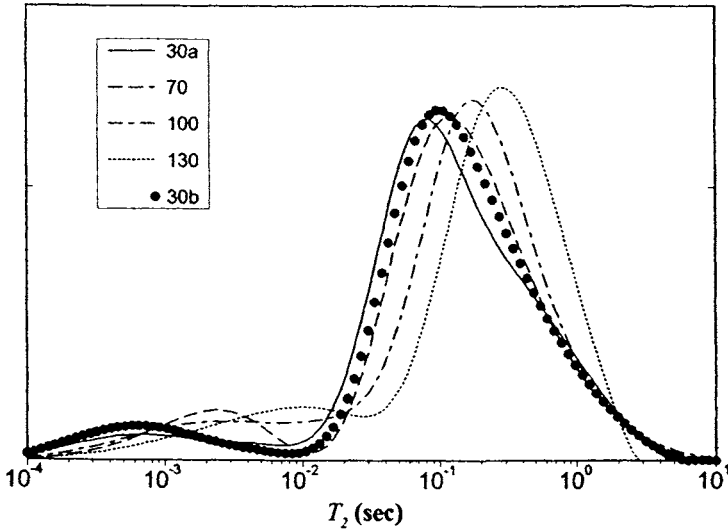


Figure 4.3: The T_2 distributions from 30°C to 130°C for a carbonate rock sample. (From Ramakrishnan et al. [184].)

dence of D_2O relaxation in porous media appears to follow that of the bulk liquid.

Relaxation of water-saturated rocks

Now that we know the relaxation of water-saturated rocks is usually in the *fast diffusion regime*, the total relaxation rate of water in the pore space is a simple volumetric rate average without involving the diffusion constant. Suppose there is a surface layer of thickness λ , such that any spin which enters this layer relaxes at the surface relaxation rate $1/T_{1m}$, where the subscript m stands for mineral or matrix. The total relaxation rate of the water in the pore space $1/T_1$ can be written as:

$$\frac{1}{T_1} = \frac{1}{T_{1B}} \left(1 - \frac{\lambda S}{V}\right) + \frac{1}{T_{1m}} \frac{\lambda S}{V}, \quad (4.7)$$

where S and V are the surface and volume of the pore space, respectively. The bulk relaxation time, T_{1B} , is of the order of 1 second, and the surface relaxation time, T_{1m} , for water in sandstones, is of the order of 0.1 to 1

millisecond. If the second term on the RHS of Eq. (4.7) is much larger than the first term, we can write:

$$\frac{1}{T_1} \approx \frac{\rho S}{V} \equiv \frac{1}{T_{1S}}, \quad \rho \equiv \frac{\lambda}{T_{1m}}, \quad (4.8)$$

where $1/T_{1S}$ is the surface relaxation rate. We now see that the total relaxation is dominated by the surface relaxation, and the relaxation time T_{1S} is proportional to the pore size a , which is characterized by V/S . Since the spins diffuse very fast in the pore space, the earlier implicit assumption of paramagnetic impurity ions being uniformly distributed over the pore surface is really irrelevant. All spins within a pore will simply relax at a monoexponential rate given by $\exp(-t/T_{1S})$. Similar arguments and Eqs. (4.7) and (4.8) also hold true for T_2 .

The self diffusion coefficient for water at 25°C is: $D = 2.5 \times 10^{-5} \text{ cm}^2/\text{sec}$. For one second, a spin can travel a distance of the order of $\sqrt{6Dt}$, which is about 120 μm . For sandstones, this is typically a distance of a few pores. Travelling through a narrow pore throat, the spin is more likely to relax because of higher frequency of hitting the surface. As a result, the signal of the total magnetization of the rock is the sum of all relaxation rates, each of which is associated with an individual pore, or at most a group of neighboring pores. Since, each relaxation time is proportional to the pore size, this becomes the basis for the assertion that NMR relaxation time distribution reflects the pore size distribution.

4.1.2 Pore size distribution

Based on the understanding in the previous section, we can describe the relaxation of the magnetization of a rock as follows:

$$M(t)/M_0 = \sum_{j=1}^n f_j e^{-t/T_j}, \quad \sum_{j=1}^n f_j = 1, \quad (4.9)$$

where $M(t)$ and M_0 represent the magnetization at time t and 0, respectively, and f_j the volume fraction of those pores which have a relaxation time T_j . Each relaxation time, $T_j = a_j/\rho$, is characterized by its own pore dimension a_j and decoupled from other T_j 's which have other values of a_j . That is, each individual pore or a group of neighboring pores, is relaxing more or less independently of other pores because of the short diffusion distance we discussed above.

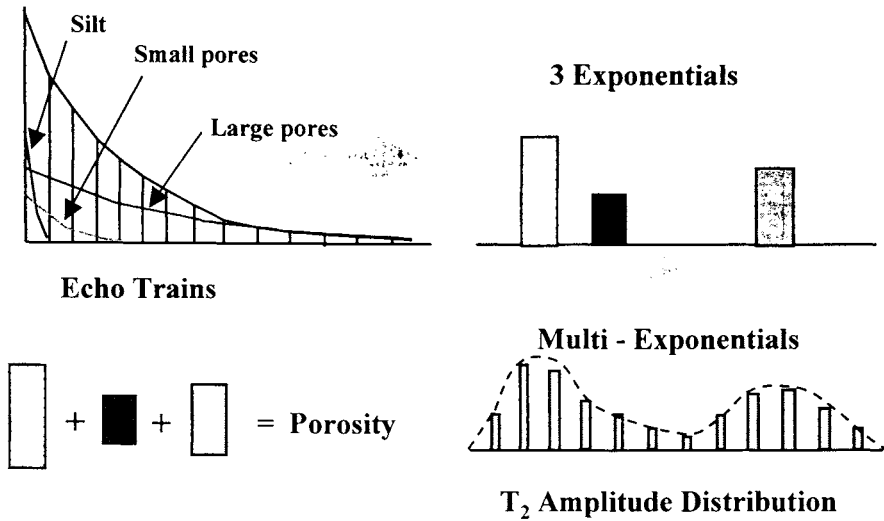


Figure 4.4: A typical analysis of water-saturated rocks with three exponential decaying curves from a nonlinear fit, as well as a continuous distribution of relaxation times equally spaced on a logarithmic scale with corresponding amplitudes from a linear inversion process.

To illustrate this, consider a possible time decaying NMR signal from a brine-saturated rock as shown in Figure 4.4. The total signal is the sum of those signals from the fluid in silt or shale, small pores and large pores, each decaying at a monoexponential rate with a relaxation time T_j which is proportional to the pore size in those regions, and with an amplitude f_j which is proportional to the amount of hydrogen nuclei (or volume fraction) in those regions.

There are several methods for obtaining the values of volume fractions f_j and relaxation times T_j . An early approach [212, 213] assumed a two- or three-exponential model as follows:

$$\begin{aligned} M(t)/M_0 &= f_A e^{-t/T_A} + f_B e^{-t/T_B} \\ 1 &= f_A + f_B, \end{aligned} \quad (4.10)$$

or

$$M(t)/M_0 = f_A e^{-t/T_A} + f_B e^{-t/T_B} + f_C e^{-t/T_C}$$

$$1 = f_A + f_B + f_C, \quad (4.11)$$

To analyze a set of data, for example $M(t_i)/M_0$, we try to minimize the following quantity

$$\chi^2 = \sum_{i=1}^m (f_A e^{-t_i/T_A} + f_B e^{-t_i/T_B} - M(t_i)/M_0)^2 \quad (4.12)$$

by choosing appropriate fitting constants f_A , f_B , T_A , and T_B . This is a nonlinear fit procedure. Sometimes, it is extended to four- or even five-exponentials, and may include also the signal at infinite time as a regression parameter. This procedure worked well for fitting most relaxation data but the interdependency of the amplitudes and time constants made it difficult to use for pore size estimation and comparisons of measurements on samples under different saturation conditions.

Another early approach, a stretched exponential fit of the following form suggested by Kenyon et al. [116, 121]:

$$M(t) = M_0 e^{(-t/T_1\alpha)^\alpha}, \quad (4.13)$$

is also a nonlinear fitting procedure. However, this method has not been used widely. It does not provide an intuitive picture of its connection to the pore size distribution. It is also difficult to incorporate other petrophysical analyses such as the T_2 cutoff for irreducible water saturation and the diffusion effect. Furthermore, there are numerous relaxation data for petrophysical samples that the stretched exponential form does not fit well.

It was soon found that these nonlinear inversion processes were not adequate for efficient processing of the large amount of NMR logging data. Therefore, instead of searching for the fitting constants, T_j , we assume a set of T_j which are equally spaced on a logarithmic time scale and fit for the amplitudes, f_j , only. The inversion procedure reduces to a linear inversion problem. Although this kind of linear inversion is well known to be ill-posed, it provides a consistent definition of irreducible water saturation using a T_2 cutoff, and it is very efficient in processing large amount of NMR log data. Also because there is a physical basis for assuming a continuous pore size distribution, this linear inversion method has gained wide acceptance in the NMR logging community. We shall discuss this inversion in more detail in Chapter 6.

Figure 4.4 also shows a *relaxation time distribution* with the relative amplitude f_j corresponding to relaxation time T_j , as compared to a previously

inverted three exponential result. Since T_j is proportional to a/ρ where a is the pore size (see Eq. (4.8)), this relaxation time distribution curve is frequently considered as a reflection of pore size distribution.

It should be noted that Eq. (4.9) was written based on the assumption that the water in the porous rock was in the fast diffusion regime. In fact, even in the slow diffusion regime, we can still analyze the time decaying magnetization using a sum of exponentials just like Eq. (4.9), except now the relaxation time T_j is proportional to a^2/D (instead of a/ρ) which is no longer proportional to the pore size a and becomes dependent on the diffusion coefficient of water.

Surface relaxivity

To treat the relaxation time distribution as a reflection of pore size distribution, not only do we have to be in the fast diffusion regime, but also the surface relaxivity, or the average surface density of paramagnetic impurity ions for each pore, should be nearly the same over the whole rock.

Surprisingly, numerous experiments on brine-saturated rock samples (Kenyon et al. [117, 118, 119], Straley et al. [208], Loren and Robinson [146]) have shown that indeed the relaxation time distribution curve is very similar to the pore size distribution curve obtained by other techniques such as mercury injection. Naturally, such similarity is not universal, as the NMR is a measure of the size of the pore bodies, whereas the mercury porosimetry is a measure of the size of the pore throats. However, if there is a reasonably consistent factor between the pore throat and the pore body, then the NMR relaxation time distribution can be quite similar to the pore size distribution determined by mercury porosimetry. Of course, if the average surface relaxivity ρ is not fairly constant over the whole scale, then the similarity between the two distributions can be affected.

Figure 4.5 from Straley et al. [208] shows an example of the T_2 distributions of four brine-saturated sandstones versus the corresponding differential mercury injection curves. The similarity between the two data sets is remarkable. Figure 4.6 from Morriss et al. [168] shows a different example for a particular type of rock, i.e., diatomite¹. The match between the two

¹A type of rock formed from the siliceous shells of diatoms which are microscopic plants living in the ocean. Diatomite's mineral composition is mainly amorphous silica (opal A). It has extremely small pore sizes.

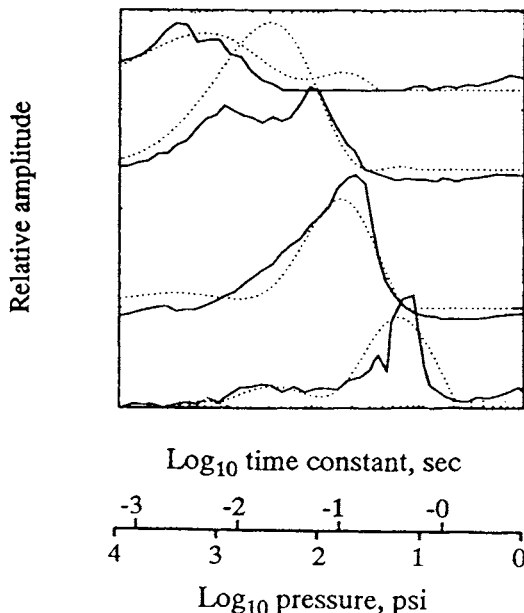


Figure 4.5: Differential mercury injection curves (solid) with the T_2 distributions (dotted) for four sandstones. (From Straley et al. [208].)

distributions is again very good until the formation of opal CT² which has a mercury injection pressure of 30,000 to 40,000 psi, and a corresponding T_2 near 0.4 ms, which the NMR T_2 measurements have completely missed.

By properly comparing the $T_{1,2}$ relaxation time scale

$$\frac{1}{T_{1,2}} = \rho_{1,2} \frac{S}{V} = \rho_{1,2} \frac{2}{r} \quad (4.14)$$

with the mercury entry pressure P

$$P = \frac{2\gamma \cos \theta}{r}, \quad (4.15)$$

assuming a cylindrical pore of radius r , where θ is the contact angle (140°) and γ is the surface tension (7.15×10^{-3} psi · cm) for mercury, we get

$$\rho_{1,2} = \frac{\gamma \cos \theta}{T_{1,2} P}. \quad (4.16)$$

²Under moderate pressures and temperatures, diatomite (opal A) becomes a partially crystallized less hydrous silica, called opal CT, before finally turning into quartz at even higher pressures and temperatures. Opal CT has even smaller pore sizes than opal A.

Thus, the surface relaxivity, based on a cylindrical model, can be easily estimated.

For Straley et al.'s example shown in Figure 4.5, the match of the two scales was found at $T_2P = 5.33 \text{ s} \cdot \text{psi}$, leading to $\rho_2 = 10.3 \mu\text{m/s}$. For Morriss et al.'s example in Figure 4.6, we have $T_2P = 10 \text{ s} \cdot \text{psi}$ and $\rho_2 = 5.5 \mu\text{m/s}$.

Kleinberg [125] found that the surface relaxation strength, determined from comparing T_2 with the mercury injection curve, is very close to that determined by comparing T_2 with the air-water capillary curve. This consistency thus lends support to the validity of the general approach. He found that ρ values for many sandstones are quite consistent with the surface relaxivity found for other porous media. Other work by Borgia et al. [32] and Bryar et al. [46] have found similar ρ_1 values for sandstones. However, Bryar et al. also found some silica gel with ρ_1 as low as $1.2 \times 10^{-3} \mu\text{m/s}$. The values of surface relaxivity for various porous materials are shown in Table 4.1.

Earlier work by Howard et al. [104], using an optical method to evaluate the pore dimension, had come up with ρ values about an order of magnitude higher, apparently due to the complication of not being able to properly characterize the surface to volume ratio of the porous media. There is a more sophisticated method which extracts the surface to volume ratio from the short time behavior of the restricted diffusion coefficient $D(t)/D_0$. We shall discuss this approach in detail in Chapter 7. Using this method, Hürlimann et al. [108] found $\rho_1 = 16 \mu\text{m/s}$ for a Fontainebleau sandstone sample.

T_1/T_2 ratio

Early laboratory NMR work on rocks was confined mostly to T_1 measurements at proton Larmor frequencies of 10 MHz and above, except for work on the NML project at Chevron in the 1950's, which covered fields corresponding to the frequency range from 8 kHz to 30 MHz. This is because the T_2 measurements are often affected by the diffusion effect caused by the internal field gradients which are difficult to characterize. In the subsequent development of pulsed NMR logging tools, it became apparent that only T_2 CPMG type measurements could offer short enough measurement time and practical logging speed. With a very short echo spacing, presumably the diffusion effect due to unknown internal field gradients in the T_2 CPMG measurements of the logging tools (Larmor frequency at 1 to 2 MHz) can be kept to a minimum.

It then becomes desirable to learn what the T_1 distribution of the rock

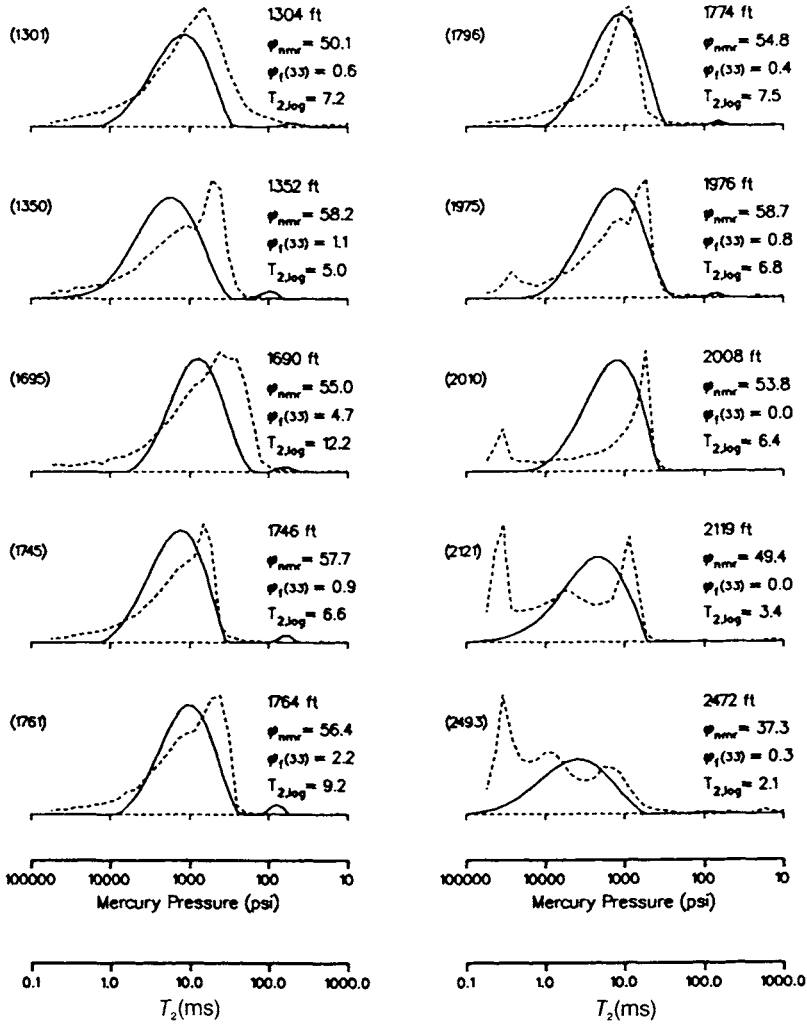


Figure 4.6: Differential capillary pressure curves (dashed) compared to T_2 distributions (solid) on cleaned and water-saturated samples from diatomite reservoir. Sample depths for the capillary pressure measurements are shown on the left in parentheses. Sample depths for the NMR measurements are shown on the right, along with NMR porosity (ϕ_{NMR}), free fluid porosity ($\phi_f(33)$), and logarithmic mean of T_2 relaxation times ($T_{2,log}$). The opal CT, characterized by a peak near 30,000 psi mercury injection pressure on the right panel of the figure, was not detected by the NMR T_2 measurements. (From Morriss et al. [169].)

Table 4.1: Surface relaxivities of several porous materials

Material	$\rho_1 (\mu\text{m/s})$	Ref.
Quartz	3	[194]
Silica gel	1-13	[217]
Porcelain	4	[146]
Alumina	3	[35]
Glass beads	5	[35]
Glass beads	11	[207]
Ultraclean crushed SiO_2	1.5^a	[42]
Microporous porcelain	2.5^a	[33]
Sandstones	2.39^b	[32]
Sandstones	0.37^b	[97]
Sandstones	0.61^c	[195]
Silica gel	1.2×10^{-3}	[46]
Porous silica glass	1.8×10^{-3}	[72]
Silica gel	4.2×10^{-3}	[91]
Silica gel	$(3.1 - 7.7) \times 10^{-3}$	[101]
Quartz sand	0.013 ± 0.015	[46]
Quartz	0.83	[122]
Synthetic Ca silicate	$0.406 - 4.04$	[85]
Silica sand	$2.89 - 3.06$	[101]
Sandstones	9.0-46	[186]
Fontainebleau sandstone	16	[108]

^acomputed with monoexponential.^bcomputed by initial rate.^ccomputed by geometric mean of relaxation rates.

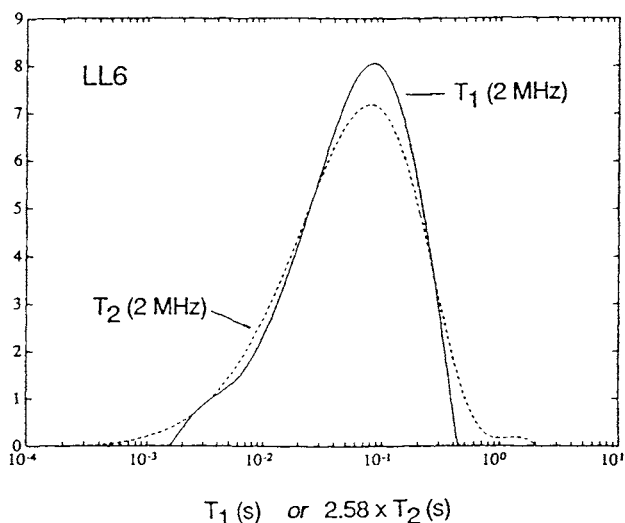


Figure 4.7: An example of T_1/T_2 cross correlation. The T_2 distribution has been shifted by a factor of 2.58 to the right to most closely match the T_1 distribution. (From Kleinberg et al. [127].)

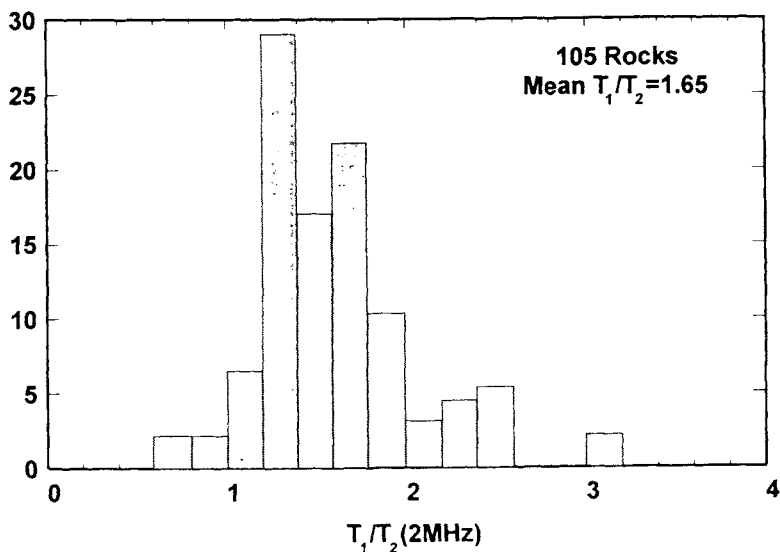


Figure 4.8: Histogram of 2 MHz T_1/T_2 ratio for 105 rocks. The CPMG echo spacing is 0.16 ms. (From Kleinberg et al. [127].)

looks like at the same Larmor frequency (1 to 2 MHz) at which we measured the T_2 distribution, what the differences between the two are, and what causes those differences.

Kleinberg et al. [126, 127] investigated many rock samples and found that when sufficiently small echo spacing is used, the diffusion effect for T_2 at 1 to 2 MHz Larmor frequency can be minimized, and that the T_2 distribution has a similar appearance to that of T_1 with a slight shift of the scale. Figure 4.7 shows such a comparison where the T_2 scale is shifted to longer times by multiplying by a factor of 2.58. Figure 4.8 shows the result of such an investigation for 105 rocks at a Larmor frequency of 2 MHz. The mean T_1/T_2 ratio is about 1.65.

The T_2 relaxation process in rocks is controlled by a number of mechanisms. Bulk relaxation can often be neglected. The surface relaxation mechanism is the same as the one that dominates the T_1 relaxation process in rocks, and provides the link to the pore size distribution of rocks. The third mechanism is molecular diffusion in macroscopic and microscopic field gradients. The CPMG pulse sequence and low Larmor frequencies can be used to minimize this diffusion effect. However, the internal gradients, induced by magnetic susceptibility contrasts between the pore fluid and solid matrix, occur at the pore scale and are difficult to eliminate completely by the CPMG pulse sequence. This kind of internal field gradient due to magnetic susceptibility contrast is proportional to the applied field strength. The enhanced relaxation rate it brings about is proportional to the square of the applied field strength. It is the primary source for the difference between T_1 and T_2 distributions [19]. As the Larmor frequency increases, the T_1/T_2 ratio is expected to increase.

Kleinberg et al.'s work [126, 127] has shown that aside from special circumstances such as the existence of extremely large internal field gradients, T_2 distribution can reflect the pore size distribution of rocks with limitations similar to those in the case of T_1 distribution. Their work also provides the important information that the T_1/T_2 ratio at 2 MHz is roughly around 1.65. This information can be used for porosity corrections in data inversion process when the repeat delay time between excitations during logging is not sufficient to have a complete polarization recovery of the spins.

4.1.3 NMR behavior of different rock types

Clastics

The T_1 or T_2 distributions of sedimentary rocks have been studied extensively. In such rocks, consistent NMR behaviors have been observed and correlated to petrophysical parameters of interest to the petroleum industry. Laboratory measurements [127] have shown that sandstones typically have a paramagnetic ion content of 1% and grain susceptibilities $\sim 10^{-5}$ cgs/cm³. The surface relaxivity typically falls in a narrow range of 1-10 $\mu\text{m/s}$ [125, 32]. The T_2 distribution is often highly correlated to the pore size distribution. When the grains are well sorted, the T_2 distribution can display a single peak at long relaxation times. Typically, a short relaxation time component is present as a shoulder or a separate bump indicating the presence of fine silt with capillary bound water or clays. The permeability formula developed from laboratory measurements usually works quite well for predicting permeability from NMR logs.

Carbonates

Carbonates usually have a surface relaxivity almost an order of magnitude smaller than that of sandstones. As a result, the diffusion coupling between pores is stronger. Thus, different pore scales, which would have different relaxation times in sandstones, are coupled to each other through the diffusion process, and merge to yield an intermediate state, or in the extreme case, a single relaxation time characteristic of the combined surface to volume ratio. Hence, the T_2 distribution of a carbonate may not reflect the pore size distribution of the rock.

Ramakrishnan et al. [183, 184], using a random walk simulation, were able to demonstrate that, indeed, the surface relaxivity plays an important role in determining whether the NMR relaxation time distribution reflects the pore size distribution. If the surface relaxivity is weak, spins diffuse easily between large and small pores, there can be a strong coupling of the decay of the magnetizations for large and small pores, and the NMR relaxation time distribution no longer reflects the pore size distribution.

They suggested a model for the analysis of carbonates which consist of a three-dimensional array of spherical grains surrounded by intergranular pore space, where all spherical grains are microporous, with a porosity of ϕ_μ . The total porosity of the system is ϕ , which is the sum of the intergranular pore

space and the micropores of the grains. Assuming that the surface relaxivity ρ is the same for both the fluid/solid interfaces in the intergranular pores and the microporous pores, they were able to show that the apparent relaxivity ρ_a of the fluid/solid interface in the intergranular pores is given by the following expression

$$\rho_a \approx \rho(1 - \phi_\mu) + \sqrt{\frac{\phi_\mu D_0}{F_\mu T_{2\mu}}} \approx \sqrt{\frac{\phi_\mu D_0}{F_\mu T_{2\mu}}}, \quad (4.17)$$

where D_0 is the diffusion coefficient of the pore fluid, F_μ is the local formation factor for the microporous grains, and $T_{2\mu}$ is the local T_2 relaxation time for the micropores (i.e., $1/T_{2\mu} = \rho S_\mu/V_\mu$, S_μ and V_μ being the surface and pore volume of the microporous medium). The second term is much larger than the first term, hence, the apparent relaxivity can be approximated by the second term.

With this model, the apparent relaxivity becomes temperature sensitive through the diffusion coefficient. They argued that this explains the temperature dependent behavior of the T_2 distribution for carbonates of grain-supported lithology when two pore length scales exist. For carbonates with mud supported lithology which also show strong temperature dependence in their T_2 distributions, they speculated that the intrinsic relaxivity ρ itself may be temperature dependent [86].

Thus, the temperature dependent behavior of the T_2 distributions for carbonates has not been fully understood yet. Some of them may be simply due to the fact that the temperature sensitive bulk relaxation time cannot be neglected for carbonates which have large pore sizes and a small surface relaxivity. One of the consequences of the temperature dependent behavior of the T_2 distributions is that the T_2 cutoff (see Section 4.3.2) determined at room temperature in the laboratory may need modifications in the analysis of the T_2 data acquired at the reservoir temperature.

Clay

The short relaxation time components in a T_2 distribution of a rock is frequently attributed to the water in clays. It is common practice to categorize the T_2 components below 3 ms as being associated with clay bound water, and those between 3 and 33 ms to be irreducible capillary bound water, and those above 33 ms to be free fluid. Clearly, this is an oversimplified partition. Earlier, Straley et al. [208] compared the clay-bound water vol-

ume (computed from cation exchange capacity/pore volume, Q_{ve} , which was determined from electrical membrane measurements) with the NMR T_2 measurements at 2 MHz, for 45 oilfield sandstones, and determined that the T_2 cutoff at 3 ms provided the best correlation between the two. Later, Prammer et al. [177] reported measurements of T_2 relaxation times for various types of clay, and showed that these clays cover a wide range of relaxation times. Their results (shown in Table 4.2) indicate that kaolinite has T_2 relaxation time peaks varying from 8 to 16 ms, whereas illite has T_2 peaks around 1 to 2 ms and montmorillonite has T_2 peaks from 0.3 to 1 ms.

Prammer et al.'s work [177], though pointing out that the convenient T_2 cutoff at 3 ms for categorizing clay is somewhat arbitrary, does suggest the possibility that the T_2 peak position can be used for clay typing.

Further work by Matteson et al. [156] suggested that such clay typing may be difficult. They conducted T_1 and T_2 measurements of brine mixed with four clays (kaolinite, smectite, illite, and glauconite) as function of compaction, at a Larmor frequency of 2 MHz and an echo spacing of 0.16 ms (similar to those used in Straley et al.'s experiments [208]). The measurements were started on clays in a state of slurry and proceeded to clays subjected to a compaction pressure of 16,000 psi. Their results (shown in Figure 4.9) indicate that there is only one relaxation peak, regardless of clay type, which shifts from over 10 ms continuously to below 1 ms as the compaction pressure increases. Thus, they concluded that it is not possible to do clay typing from the NMR T_2 relaxation time distributions if the compaction history is not known.

From their experiments, the shift of the T_2 relaxation time appears to be simply reflecting the change of pore volume (surface area, presumably, should stay constant) as a function of the compaction. Assuming the surface area of the clay stays the same during compaction, the computed surface relaxivity remains nearly constant as a function of compaction. Among the four clays they studied, kaolinite has the lowest value for the surface relaxivity, 1.8 $\mu\text{m/s}$, and glauconite has the highest, 3.3 $\mu\text{m/s}$. The T_2 peak positions of the slurry and centrifuged samples are, from the longest to shortest relaxation time, kaolinite, smectite, illite, and glauconite. For compacted samples from 500 to 16,000 psi, the kaolinite has the longest relaxation time, whereas smectite, illite, and glauconite have the same peak position which is shorter than that of kaolinite.

One of Matteson et al.'s findings was that there does not exist a separate peak as clay-bound water. All the observed peaks appear to be simply due

Table 4.2: Prammer et al.'s laboratory T_2 measurements at 1 MHz and at 25°C of various types of clay. Samples were saturated with different amounts of brine, pressed at 2,500 psi, and sealed. Water weight was determined from weight loss by overnight drying in a vacuum chamber at 103°C. The apparent transverse surface relaxivities ρ_2 are calculated from specific surface area, the water-to-dry weight ratios, and T_2 values. All measurements were performed at an echo spacing of 0.5 ms.

Clay Type	Water Wt ^a (%)	Signal ^b (%)	T_2^c (ms)	ρ_2^d ($\mu\text{m/s}$)	Remarks
Montmorillonite	7.0	20	-	-	$T_2 < 0.2\text{ms}$
	18.9	90	0.3	1	
	31.1	100	0.5	1	
	54.4	100	1	0.9	$T_1 = 1.5\text{ms}$
Illite	8.8	90	1	0.9	
	15.8	100	2	0.8	
Kaolinite	11.7	100	8	0.8	
	17.4	100	12	0.8	
	20.0	100	16	0.7	$T_1 = 30\text{ms}$
Chlorite	7.5	100	5	0.4	ρ_2 too low ^e

^aWater weight per dry clay weight.

^bRatio of measured NMR amplitude per milliliter water content.

^c T_2 of clay bound water.

^dApparent transverse surface relaxivity.

^e ρ_2 may be too low as a result of overestimated surface area.

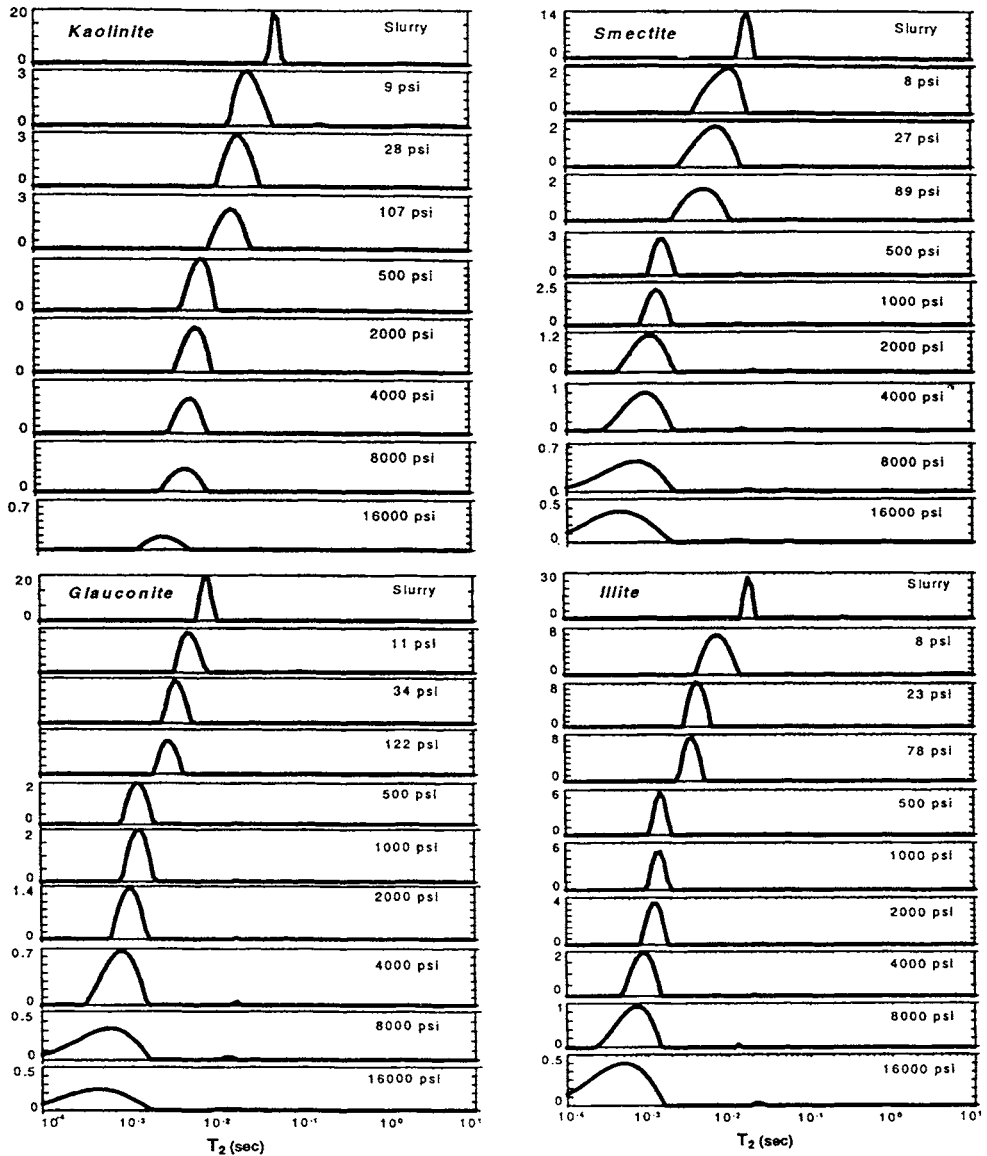


Figure 4.9: The T_1 and T_2 distributions of clay-brine slurries for kaolinite, smectite, illite, and glauconite. (From Matteson et al. [156].)

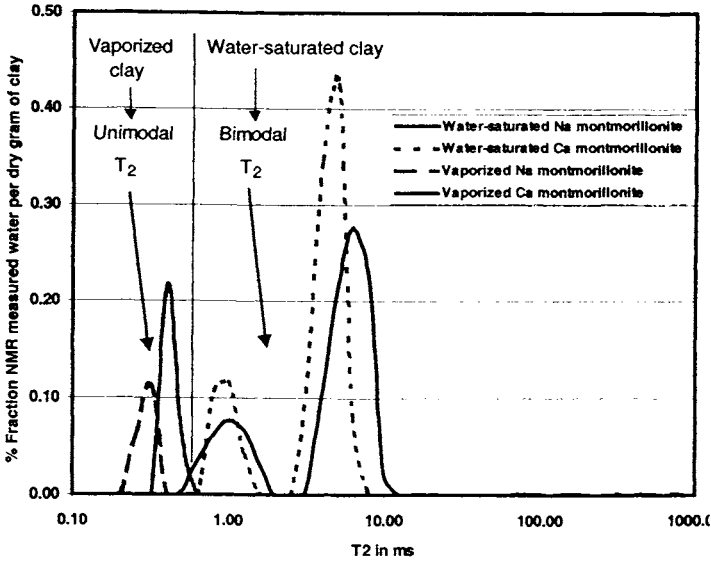


Figure 4.10: Typical T_2 distributions of hydrated montmorillonite. (From Chitale et al. [61].)

to the pore water between the clay particles. Chitale et al. [61], on the other hand, found that for hydrated samples of Ca and Na montmorillonite there exists a fast T_2 component clustered around 1 ms. As the water content increases beyond the critical saturation for the clay, the T_2 distribution becomes bi-modal, with the fast component staying the same near 1 ms and the slow component moving toward longer time. Chitale et al.'s result is shown in Figure 4.10. While there are differences between the experimental findings, it seems fair to say that kaolinite has the longest relaxation time under the same compaction history. Thus, at least distinguishing kaolinite from other clays appears possible.

4.2 Diffusion in porous media

Near the end of Chapter 2, we discussed the effect of diffusion of spins in a constant field gradient in an infinite medium. In a Hahn's echo measurement, the spin dephasing due to diffusion causes an additional magnetization decay besides the normal T_2 relaxation. The total decay is given by the following

expression

$$\exp\left(-\frac{2\tau}{T_2} - \frac{2}{3}\gamma^2 G^2 D\tau^3\right), \quad (4.18)$$

where 2τ is the echo spacing, γ is the gyromagnetic ratio, G the constant field gradient, and D the self diffusion coefficient of the molecule where the spin resides. For a CPMG type measurement, the magnetization decay of the n th echo is given by the following expression

$$\exp\left(-\frac{2n\tau}{T_2} - \frac{2}{3}\gamma^2 G^2 Dn\tau^3\right). \quad (4.19)$$

Diffusion only affects T_2 , not the T_1 measurement³, because there are spin dephasing and refocusing involved in the former.

We should first realize that, if the field is perfectly homogeneous everywhere, the diffusion of spins will not produce any degradation to the magnetization. This is because all spins, no matter where they go, precess at the same Larmor frequency, and there is no dephasing. If there is a field gradient and no diffusion, then the dephased spins can be brought back in phase by a π pulse for refocusing in a spin echo experiment. In reality, neither a perfectly homogeneous field nor a non-diffusive spin is possible. Thus, when both field gradient and diffusion are present, dephasing will occur and cause further degradation in the T_2 magnetization decay. Eqs. (4.18) and (4.19), which describe these decays, are strictly valid only for an infinite fluid medium, and are only approximately correct in a porous medium under conditions of restricted diffusion.

4.2.1 Enhanced T_2 relaxation

As we move from relaxation in an infinite fluid medium to relaxation in a fluid-saturated porous medium, analysis becomes more complicated. However, as long as the diffusion time is short, i.e., the τ in Eq. (4.19) is small, we can assume most of the spins do not experience the presence of the pore wall. Thus, Eq. (4.19) derived for an infinite uniform fluid is still approximately correct for the pore fluid in the rock. We can write the measured apparent T_2 relaxation rate of the rock in the following form:

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \frac{1}{T_{2S}} + \frac{1}{T_{2D}} \quad (4.20)$$

³See footnote [1] of Chapter 2 in p. 29

where the $1/T_2$ in Eq. (4.19) is replaced by $1/T_{2B} + 1/T_{2S}$, and the enhanced relaxation rate is replaced by $1/T_{2D}$.

T_{2B} represents the T_2 value for the bulk fluid, and T_{2S} represents the relaxation caused by the surface relaxivity which we discussed in the previous section, i.e.,

$$\frac{1}{T_{2S}} = \rho_2 \frac{S}{V}. \quad (4.21)$$

The enhanced T_2 relaxation rate, caused by the effect of spin diffusion in a field gradient, is given by:

$$\frac{1}{T_{2D}} \equiv \frac{(\gamma G \tau)^2 D}{3} = \frac{(\gamma G t_E)^2 D}{12}, \quad (4.22)$$

where $t_E = 2\tau$ (TE) is the time between echoes.

In general the value of $1/T_{2B}$ is much smaller than $1/T_{2S}$, and usually can be neglected. Then the relaxation rate $1/T_{2S}$ is enhanced by the presence of $1/T_{2D}$. For porous systems such as rocks, which have multiple pore sizes, T_{2S} is replaced by a distribution of T_2 values representing different pore sizes, and each of them is modified by a constant enhanced relaxation rate $1/T_{2D}$ when the rock is placed in an externally applied constant field gradient and there are no significant internal gradients. Thus, a large T_{2S} experiences a large shift in the apparent T_2 , whereas a small T_{2S} acquires only a very small shift in its apparent value.

As we stated earlier, Eq. (4.20) is approximately valid for fluid-saturated rocks as long as the diffusion time (i.e., TE or 2τ) is very short. For the same length of diffusion time, a large field gradient causes much faster dephasing. Naturally, the length of this diffusion time, for which Eq. (4.20) is approximately valid, depends on the value of the externally applied constant field gradient. The larger the value of the field gradient, the smaller the value of the diffusion time for which Eq. (4.20) is approximately valid.

Under such constraints, Eq. (4.20) is frequently used for the analysis of T_2 distributions of rocks in a well-defined externally applied constant field gradient. Note that $1/T_{2D}$ cannot properly characterize the enhanced T_2 relaxation rate caused by the internal local field gradients within the pore space, because the latter are not uniform at the pore scale.

Thus, when the diffusion time is long, there are at least two additional considerations we need to think about in going from the infinite uniform fluid to a fluid-saturated porous medium, i.e., internal field gradients and

restricted diffusion. These two issues are very complicated and difficult to treat theoretically in a rigorous manner. We shall study these in more detail in Chapter 7. For now, we will only review these qualitatively, and discuss the general approximate approaches by making several assumptions which the logging environment may or may not satisfy.

4.2.2 Restricted diffusion

As the spins diffuse in the pore space, they encounter the pore wall of the solid grains and redirect their random walk path to other directions in the pore space. Certainly, the diffusion profile will differ from the profile of spins which are in an infinite medium and are not restricted by the pore geometry. As a result, the spin echo amplitude, i.e., Eqs. (4.18) and (4.19), will be modified accordingly. Since these equations are frequently used in NMR log interpretation for estimating the diffusion coefficient of the oils or the field gradients in the rocks, it is important to learn what needs to be modified and what are the bounds of validity for these equations.

At times short enough that most spins do not encounter the pore walls or experience a significant change in local gradient, we expect Eqs. (4.18) and (4.19) to remain valid. As time goes on, since diffusion is restricted, we expect the effective diffusion coefficient $D(t)$ to decrease with time. In the short time regime, only spins within a distance $\sqrt{D_0 t}$ from the pore walls, of area S , sense the presence of the walls. Thus, only a fraction $\sqrt{D_0 t} S / V_p$ of the total spin population, where V_p is the pore volume, deviate from their diffusive behavior in the free medium. Thus the diffusion behavior is expected to deviate from its free medium behavior by the same order of magnitude. Mitra et al. [165, 166, 167], using an approximate solution for the diffusion propagator near a pore wall, were able to derive the following temporal behavior for restricted diffusion in the short time regime:

$$\frac{D(t)}{D_0} = 1 - \frac{4}{9\pi} \frac{S}{V_p} \sqrt{D_0 t} - \frac{S}{12V} \left\langle \frac{1}{R_1} + \frac{1}{R_2} \right\rangle D_0 t + \frac{1}{6} \frac{\rho S}{V_p} D_0 t + \mathcal{O}[(D_0 t)^{3/2}] \quad (4.23)$$

This relation was later used by Hürlimann et al. [108] and Slijkerman and Hofman [202] to obtain the surface relaxivity experimentally.

For a long diffusion time, the time dependent diffusion constant approaches a constant value if the pore space is connected. If the surface relaxivity ρ is zero, then the asymptotic value of the effective diffusion coefficient is related to the electrical conductivity via the standard Einstein

relation [75]

$$\lim_{t \rightarrow \infty} \frac{D(t)}{D_0} = \frac{1}{F\phi} \quad (4.24)$$

where $F \equiv \sigma_p/\sigma_e$ is the formation factor⁴ of the porous medium with σ_p being the conductivity of the pore fluid (brine) and σ_e being the effective conductivity of the porous medium, assuming the rock matrix is non-conductive. When $\rho \neq 0$, the infinite time limit of $D(t)/D_0$ is related to another electrical conductivity problem, where the pore fluid has a modified electrical conductivity – see Section 7.1.2, Eq. (7.35), and Ref. [21].

4.2.3 Spin echo amplitude

To derive the T_2 spin echo amplitude in a restricted geometry, one has to evaluate the phase shift accumulated by a precessing spin in a Hahn spin echo experiment:

$$\Phi = -\gamma \left(\int_0^\tau dt \delta H[\mathbf{r}(t)] - \int_\tau^{2\tau} dt \delta H[\mathbf{r}(t)] \right) \quad (4.25)$$

where δH is the local field deviation from a uniform background field H_0

$$H(\mathbf{r}) = H_0 + \delta H. \quad (4.26)$$

The average spin echo amplitude is evaluated by

$$\langle e^{-i\Phi} \rangle = \frac{1}{V_p} \int \int d\mathbf{r} d\mathbf{r}' G(\mathbf{r}\mathbf{r}'|t) e^{-i\Phi} \quad (4.27)$$

which takes into account all possible paths of a diffusing spin by means of the diffusion propagator $G(\mathbf{r}\mathbf{r}'|t)$ and averages over all spins in the pore space. The diffusion propagator $G(\mathbf{r}\mathbf{r}'|t)$ is the solution of the following diffusion equation:

$$\begin{aligned} D\nabla^2 G(\mathbf{r}\mathbf{r}'|t) &= \frac{\partial G(\mathbf{r}\mathbf{r}'|t)}{\partial t} && \text{in the pore space} \\ \hat{\mathbf{n}} \cdot D\nabla G(\mathbf{r}\mathbf{r}'|t) + \rho G(\mathbf{r}\mathbf{r}'|t) \Big|_S &= 0 && \text{at the interface} \\ G(\mathbf{r}\mathbf{r}', t=0) &= \delta(\mathbf{r} - \mathbf{r}') && \end{aligned} \quad (4.28)$$

⁴Formation factor is a quantity frequently used in petrophysics, and it is related to another frequently used petrophysical quantity, tortuosity α , through the relation $\alpha = F\phi$, where ϕ is the porosity.

Neuman [171] carried out an approximate version of this calculation for the simple geometries of parallel plates, sphere, and cylinder in a constant field gradient, by evaluating $\langle e^{-i\Phi} \rangle$ under the assumption that the phase Φ is distributed as a Gaussian. He showed that when τ (half of the echo spacing) is much less than a^2/D , i.e., the diffusion time across a pore dimension a , the magnetization reduces to the same form as in the case of the infinite medium:

$$M(2\tau) \sim \exp\left(-\frac{2\tau}{T_2} - \frac{2}{3}\gamma^2 G^2 D \tau^3\right) \quad (4.29)$$

This is what we expected. In fact, in a restricted geometry such as porous media like rocks, we have to satisfy two requirements before we can safely use Eq. (4.29). First, the distribution of the phase (Eq. (4.25)) should be Gaussian or nearly Gaussian. The Gaussian approximation is good only for a limited range of diffusion time and field gradient strength. Beyond that, the distribution is no longer Gaussian, and an equation of the form of Eq. (4.29) is no longer valid. Second, when the Gaussian approximation is valid and a magnetization decay form of Eq. (4.29) is used, we need to replace D with a time dependent diffusion coefficient $D(t)$ when the diffusion time t is long. As this depends on many factors, such as surface relaxivity, pore size, field strength, etc., it is difficult to set a guideline. However, it is not unusual that with an echo spacing of 2 ms, the diffusion coefficient may experience a 10 to 20% reduction due to restricted diffusion.

A more rigorous discussion of this problem appears in Section 7.1.3.

4.2.4 Internal gradients

The internal field gradients of porous media are induced mainly by the magnetic susceptibility contrast between the solid matrix and pore fluid. The magnitudes of these internal gradients depend not only on the magnetic susceptibility contrast but also on the pore geometries. The local field gradient can be extremely high near the sharp corners of solid grains. When spins diffuse near those corners, they can be quickly relaxed.

Strong internal field gradients present at least two problems. First, T_2 of the short relaxation time components can relax so fast that the signal is not captured, leading to an underestimated porosity. Second, the shift of the T_2 peaks towards shorter relaxation times can cause errors in the computation of the diffusion coefficient or viscosity of the oils, or other errors in the in-

terpretation of the T_2 distributions. Thus, it is important to understand the effect of strong internal field gradients and further to quantify such an effect.

There have been numerous attempts to quantify such internal field gradient distribution (Bendel [12], Hürlimann [109], Appel et al. [10], Zhang et al. [231]). But the complexity of the problem makes rigorous analysis of experimental data extremely difficult. We attempt to express the CPMG measurement result for a fluid-saturated rock with internal field gradients as follows:

$$\frac{M(t_i)}{M_0} = \sum_j f_j \exp\left[-\frac{t_i}{T_j}\right] \int dg P_j(g) \exp\left[-\frac{1}{3}\gamma^2 g^2 D_p \tau^2 t_i\right] \quad (4.30)$$

where $t_i = 2i\tau$, j is referring to different pore sizes, f_j is the volume fraction of pore size j , and the integration is over different possible values of the field gradient g in that type of pore, with the normalized field distribution $P_j(g)$, i.e.,

$$\int P_j(g) dg = 1. \quad (4.31)$$

Here, we assume the enhanced relaxation rate is always proportional to g^2 , which may not be true, and we also assume that all pores of size j have the same distribution of internal field gradients, which, of course, may be a shaky assumption as well.

The analysis of experimental data is very difficult because it is not trivial to decouple the surface relaxation term from the field gradient contribution. However, if we make the drastic assumption that $P_j(g)$ for all pore sizes are the same, the integration in Eq. (4.30) becomes a common factor for all pore sizes and can be pulled out of the summation. Then, the ratio of the magnetization measured with an arbitrary echo spacing τ to the magnetization measured with the smallest echo spacing τ_s becomes:

$$\frac{M(t_i, \tau)}{M(t_i, \tau_s)} = \int dg P(g) \exp\left[-\frac{1}{3}\gamma^2 g^2 D_p \tau^2 t_i\right] \quad (4.32)$$

where we assume

$$M(t_i, \tau_s) \longrightarrow \sum_j f_j \exp\left[-\frac{t_i}{T_j}\right] \quad (4.33)$$

and

$$\int dg P(g) \exp\left[-\frac{1}{3}\gamma^2 g^2 D_p \tau_s^2 t_i\right] \longrightarrow 1 \quad (4.34)$$

for the smallest τ , i.e., τ_s . Once Eq. (4.30) is reduced to the form of Eq. (4.32), it is possible to obtain the internal field gradient distribution $P(g)$ by inverting this Gaussian transformation. Obviously, many rocks do not satisfy such a requirement, and the interpretation of the distribution $P(g)$ obtained this way would then be ambiguous.

4.3 Petrophysical applications

To understand and/or interpret the results of NMR logs from borehole measurements, it is imperative to conduct NMR laboratory measurements on brine and/or oil saturated rock samples first. Such measurements can be carried out to almost any desired accuracy using various means not readily available in a logging situation. For example, the measured NMR signals can be conveniently stacked many times to obtain a very high signal to noise ratio. One can study how the level of noise tends to affect the characteristics of the T_2 distributions. When the water and oil signals are both present in the measurements, one can soak the core plugs in D_2O to remove the water signal, or use the Magic Angle Spinning (MAS) technique to differentiate the two phases in high magnetic field. The determination of irreducible water saturation, the understanding of oil T_2 relaxation times with respect to its viscosity and diffusion coefficient, as well as the establishment of permeability estimators, are all important steps in the laboratory for the understanding, interpretation, and applications of NMR logs to formation evaluation. In the following we shall discuss them in detail.

4.3.1 Porosity

The NMR signal, when extrapolated back to zero time (in a T_1 or T_2 measurement), is proportional to the population of the hydrogen nuclei in the probed zone. When properly calibrated, the NMR measurement is an excellent way to measure porosity. Figure 4.11 (from Straley [208]) shows the excellent comparison of 2 MHz NMR porosity of T_2 measurements with buoyancy porosity for 192 sandstones. However, if the T_2 measurement is conducted at very high magnetic field, the correlation can be poor. This is because the magnetic susceptibility contrast induced internal field gradients are proportional to the applied field strength, and the enhanced relaxation rate is proportional to the square of the applied field strength. Thus, as the

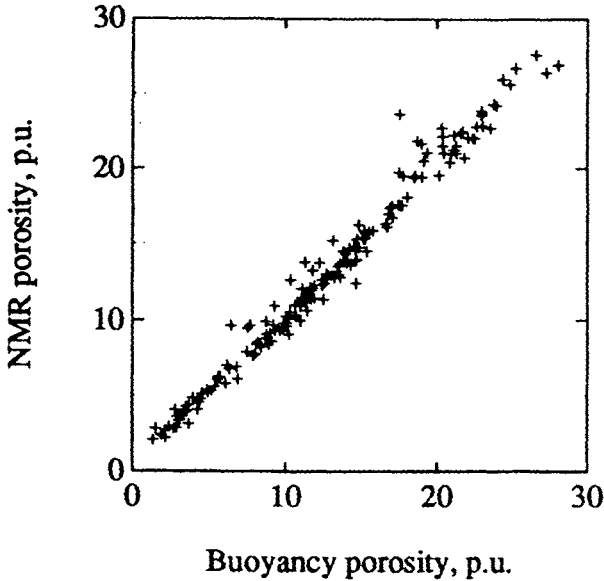


Figure 4.11: Comparison of 2 MHz NMR porosity with buoyancy porosity for 192 sandstones. (From Straley et al. [208].)

field strength increases, the increased T_2 relaxation rate causes some of the NMR signal to decay too quickly to be recorded.

Figure 4.12 shows a comparison of NMR porosity at three field strengths to buoyancy porosity. It is clear that those measured at 85 MHz suffered significant signal loss from induced internal gradients. Typically, one should do these experiments near the logging frequencies, i.e., 1 to 2 MHz. Then, with a reasonably small echo spacing, internal gradient effects can be minimized.

However, even at Larmor frequencies of 1 to 2 MHz, porosity can be underestimated when the Q of the rf coil is degraded by highly saline brine in the rock pores. This problem can be dealt with by proper calibration and adjustments to the Q of the rf coil.

4.3.2 Irreducible water saturation

One of the more important applications of NMR logging is to estimate the irreducible water saturation. Irreducible water saturation is not a precisely defined term. It can be loosely defined as the water saturation below which increases in capillary pressure will result in negligible decreases in its value.

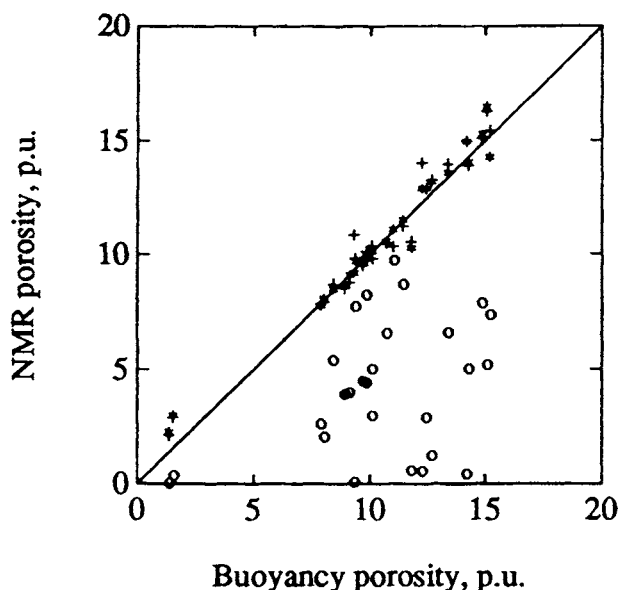


Figure 4.12: Comparison of NMR porosity of T_2 measurements at three field strengths to buoyancy porosity. o is 85 MHz, + is 2 MHz, and * is 1 MHz data. (From Straley et al. [208].)

Capillary pressure is derived from the difference in density between oil and brine which causes the oil to displace or “drain” the water from the formation. In the oil industry, a capillary pressure of 50 psi is often used to determine irreducible water saturation in poorly consolidated core. For well consolidated core, a capillary pressure of 100 psi or higher is typically used.

Early experiments by Timur [213] use a three-exponential model to analyze the T_1 relaxation data for 100% fully brine saturated core samples. The fraction associated with the long T_1 component(s) (defined as Free Fluid Index, FFI) beyond a critical T_1 value was plotted against the movable fluid determined by the core measurements with a desaturation pressure of 50 psi. A good linear correlation was found. Usually, the irreducible water saturation is found to be related to the shortest T_1 component. Thus, the NMR measurements of a brine-saturated rock provide a good estimate on its irreducible water saturation.

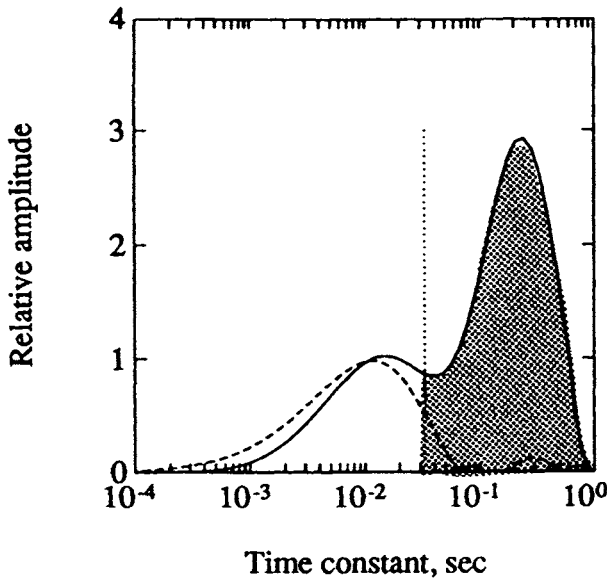


Figure 4.13: The T_2 distributions of a sandstone when it is fully saturated (solid) and when it is desaturated after centrifuging at 100 psi (dashed). The vertical cutoff in the T_2 distribution separates the porosity into the one for free fluid (shaded) and the one which is irreducible, with the latter equal to the water content of the desaturated sample. (From Straley et al. [208].)

T_2 cutoff

When a set of relaxation times equally spaced on a logarithmic scale is chosen, and a regularized inversion process is adopted for the analysis of NMR data, numerous laboratory experiments [168, 208, 76, 56, 120] indicate that a T_2 cutoff value of 33 ms for sandstones and 100 ms for carbonates are appropriate for irreducible water saturation determination. Of course, these are based on averages determined empirically from experiments. Caution should be used in the application of such cutoffs. Measurements on cores from the reservoir studied provide the most reliable cutoffs.

Figure 4.13 shows T_2 distributions of a sandstone sample when it is fully saturated and after it is desaturated by centrifuging at an equivalent air/brine capillary pressure of 100 psi. The T_2 cutoff (vertical line) separates the distribution into the porosity containing the movable fluid with long T_2 and the porosity containing the irreducible water with short T_2 . Figure 4.14 shows a cross plot of the Free Fluid Index (FFI), obtained by centrifuging a suite of

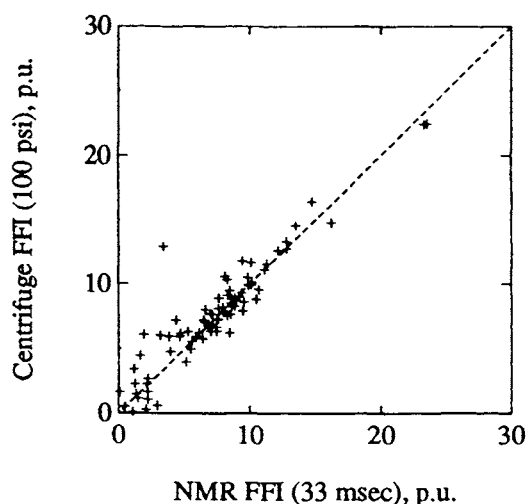


Figure 4.14: The crossplot for a suite of sandstones of centrifuge FFI at 100 psi versus the NMR determined FFI based on a T_2 cutoff at 33 ms. (From Straley et al. [208].)

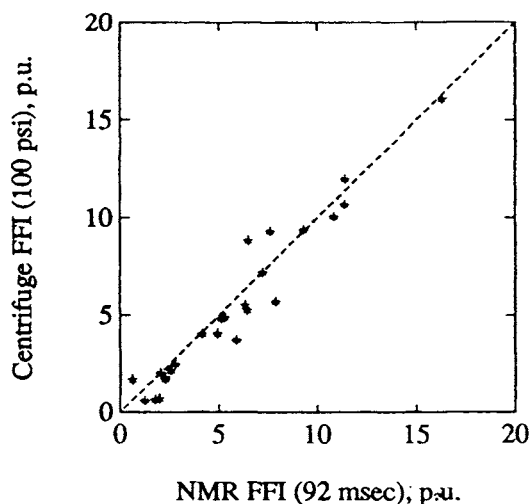


Figure 4.15: The crossplot for a suite of carbonates of centrifuge FFI at 100 psi versus the NMR determined FFI based on a T_2 cutoff at 92 ms. (From Straley et al. [208].)

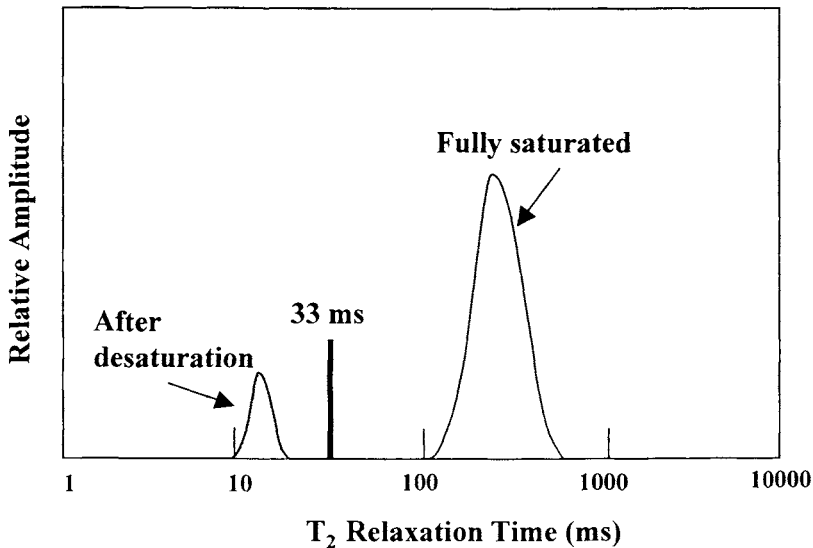


Figure 4.16: The T_2 distributions for a rock sample with a relatively uniform pore size when it is fully saturated and when it is desaturated (shaded). The T_2 peak emerges below 33 ms only after the desaturation. Note that the normal T_2 cutoff of 33 ms will not work in this case.

sandstone samples at an equivalent capillary pressure of 100 psi, versus the FFI determined by the cutoff at 33 ms on the T_2 measurements made on the same samples. Figure 4.15 shows a similar cross plot for carbonates where a T_2 cutoff of 92 ms was used. The appropriate T_2 cutoff for sandstones or carbonates can vary from reservoir to reservoir. For example, the T_2 cutoff for sandstones can vary from 20 to 50 ms, whereas that for carbonates from 90 to 150 ms. The fact that the T_2 cutoffs for carbonates are typically much larger than those for sandstones is understandably due to the much smaller surface relaxivity for carbonates.

Spectral representation

Frequently, when the pore size is uniform with little variation, the T_2 distribution for a fully water-saturated core has only one peak at long relaxation times, far above the T_2 cutoff. As the core is desaturated, the water in the pore space is emptied, leaving only small amount of water in the interstitial

space near the grain contacts. This results in the emergence of a T_2 peak at a short relaxation time which was absent before desaturation. Figure 4.16 shows T_2 distributions for such a situation. In this case, the regular cutoff will not work, and it will lead to a zero value for the irreducible water.

There have been various proposals suggested to handle a situation like this. Some use a variable cutoff for different lithologies or facies, where each cutoff is determined empirically from laboratory core measurements.

Some use what is called the Spectral BVI method (Coates et al. [64]), namely, a combination of fractions of the amplitude for all the relaxation times, to determine the irreducible water saturation. For short relaxation times, nearly all amplitudes contribute to the irreducible water. For long relaxation times, only a small portion of the amplitudes contribute to the irreducible water. These different fractions of amplitude for different relaxation times are also empirically determined from core measurements.

Bi-exponential model

Another method is to use an experimentally determined empirical relation between the irreducible water saturation, S_{wirr} and the $T_{2,LM}$ (the logarithmic mean of T_2 relaxation times) for the prediction of S_{wirr} . All these methods work equally well.

Figure 4.17 is a crossplot of S_{wirr} vs $T_{2,LM}$. The data are from the NMR Rock Catalogue from the combined companies ART/SCINTEF/RESLAB in England and Norway. The S_{wirr} were obtained by centrifugation using an equivalent air/brine capillary pressure of 100 psi. The $T_{2,LM}$ were determined at an echo spacing of 0.35 ms using a 2 MHz, homogeneous field spectrometer. The data is fitted using a bi-exponential model. The fitted equation is given by:

$$S_{wirr} = 0.6101 \exp(-T_{2,LM}/15.9) + 0.3688 \exp(-T_{2,LM}/276.8) \quad (4.35)$$

where the $T_{2,LM}$ is in ms. Long $T_{2,LM}$ values are clearly associated with small S_{wirr} values but the data density at long $T_{2,LM}$ values is sparse.

The core used in the study comes from different oil companies but is largely from North Sea reservoirs. For logging tools with a field gradient, modifications to Eq. (4.35) are needed to account for diffusion effects in the field gradient.

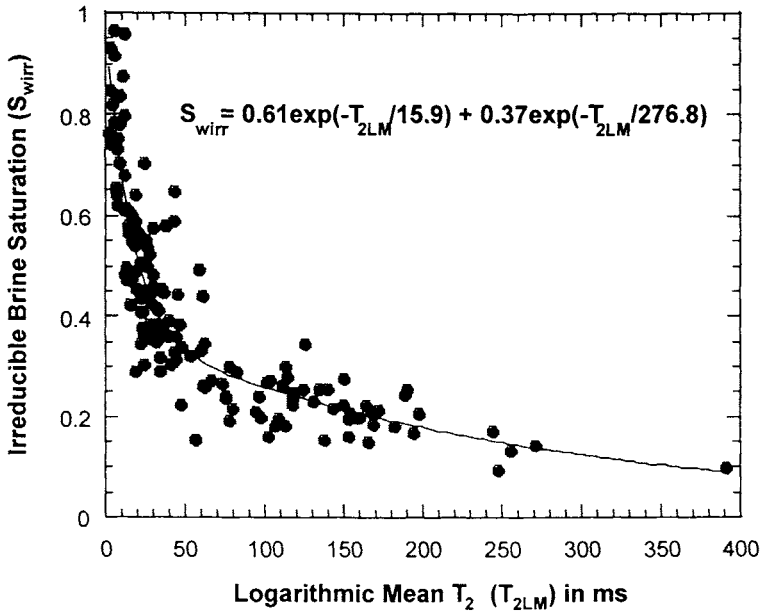


Figure 4.17: The irreducible water saturation, S_{wirr} , versus $T_{2,LM}$ of fully brine-saturated state for a suite of rock samples. The data are fitted with a two-exponential model.

Capillary curve

As we discussed earlier, the pore size distribution determined by mercury porosimetry is similar to the T_2 distribution of the rock. Similarly, its cumulative quantity, i.e., the capillary pressure curve, is connected to the cumulative T_2 distribution. This is shown in Figure 4.18. In capillary pressure based estimates of S_{wirr} , we typically use the apparent asymptotic value of S_w , beyond which very large P_c values would be needed to reduce S_w , or we determine a maximum capillary pressure for a reservoir and argue that the S_w at that pressure, in an air/brine or oil/brine drainage experiment, is the irreducible brine saturation. Such data can be interpreted in terms of a minimum pore size through which the non-wetting phase (oil) can displace the wetting phase (brine). In a similar manner, a cutoff in the T_2 distribution is often used to estimate S_{wirr} .

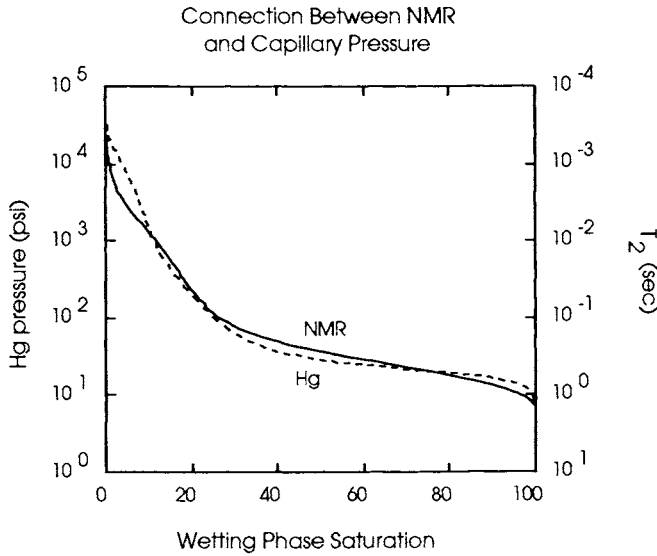


Figure 4.18: The connection between capillary pressure curve and cumulative T_2 distribution. (From Kleinberg [125].)

4.3.3 Permeability predictor

$T_{2,LM}$ equation

The fact that NMR log can be used for permeability prediction is because the T_2 relaxation time is proportional to pore size (or pore volume to surface ratio):

$$T_2 \approx \frac{V}{\rho_2 S} \quad (4.36)$$

and provides the length scale for permeability prediction. Kenyon et al. [121] and Banavar and Schwartz [11] have shown that the following equation provides a good permeability prediction for sandstones:

$$k = T_{1,LM}^2 \phi^4 \quad (4.37)$$

A general form of $k = a T_{2,LM}^b \phi^c$ can be used with the prefactor, a (which includes the $T_{1,LM}/T_{2,LM}$ ratio), and the exponents, b and c , specifically determined from fitting a set of core permeability measurements.

Other forms, such as $k = a T_{2,LM}^b / F^c$, (where F is the formation factor) have been suggested [222, 115, 113, 78, 195, 28]. Both forms work about

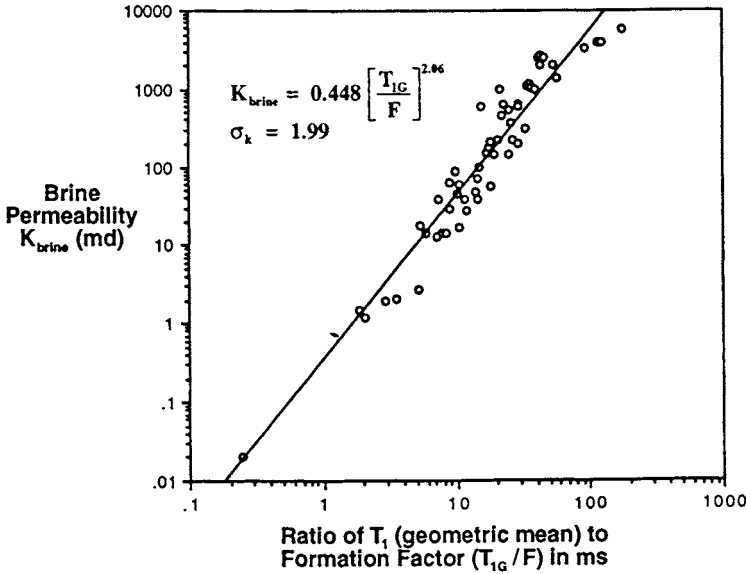


Figure 4.19: Brine permeability versus $T_{1,LM}/F$ for a suite of sandstones. The dimensionless quantity σ_k is defined in Table 4.3. (From LaTorraca et al. [132].)

equally well within experimental error. Table 4.3 [132] shows some of the comparison between different models. Figure 4.19 shows such a plot for sandstones from a single field.

This suggested form of permeability predictor has its physical basis. It was shown by Katz and Thompson [115], Johnson et al. [113], and many others, that the permeability of a porous medium can be well described by the following form:

$$k \sim L^2(\sigma/\sigma_f) = L^2/F = L^2\phi^m. \quad (4.38)$$

Here σ and σ_f are the electrical conductivities of the fully saturated rock and the pore fluid, respectively, L is a length scale characteristic of the pore geometry, and F is the formation factor. The last part of Eq. (4.38) is simply from Archie's law (i.e., $F \sim \phi^{-m}$). As pointed out earlier, T_2 is proportional to the characteristic length of the pore scale, and hence L . Naturally, different pore geometries will affect the value of the exponents for L and ϕ . They are usually empirically determined for a suite of rocks from a reservoir.

Table 4.3: Least-squares fit permeability estimators.

Reference	Permeability (mD)	σ_k^a	n^b
Kenyon et al. [121]	$K_{brine} = 1.17(T_{1\alpha}^2\phi^4)^{1.13}$	2.7	
Sen et al. [195]	$K_{brine} = 0.794(T_{1\alpha}/F)^{2.15}$	3.3	
Borgia et al. [28]	$K_{air} = 3.25(T_{1\alpha}^2/F^2)^{0.79}$	2.1	
	$K_{air} = 0.076(T_{1\alpha}/F)^{0.94}$	2.3	
	$K_{air} = 10.7(T_{1\alpha}^2\phi^4)^{0.72}$	1.9	
LaTorraca et al. [132]	$K_{brine} = 0.0063(T_{1G}/F)^{1.23}$	3.7	257
	$K_{brine} = 0.43(T_{1,LM}/F)^{2.16}$	3.4	257
	$K_{brine} = 0.33(T_{1\alpha}/F)^{2.05}$	3.4	257
	$K_{brine} = 1.21(T_{1,LM}^2\phi^4)^{1.00}$	3.6	257
	$K_{brine} = 4.00(\sum_{i=1}^3(A_i\phi)^4T_{1i}^2)^{0.933c}$	3.7	257
(Clastic subset)	$K_{brine} = 0.29(T_{1,LM}/F)^{2.16}$	2.0	57
	$K_{brine} = 0.45(T_{1,LM}/F^*)^{2.06d}$	2.0	57

* K is in mD, T_1 in ms, and ϕ in fraction.

^a σ_k is the factor by which the estimated permeability misses the measured permeability, i.e., $\sigma_k = 10^\epsilon$, where ϵ is obtained from $\log_{10} K = a + b \log_{10}(\text{predictor}) \pm \epsilon$.

^bNumber of samples.

^c A_i and T_{1i} are fitting constants in a three-exponential model fit for the T_1 distribution. Thus, $A_i\phi$ and T_{1i} are the partial porosity and the corresponding T_1 component, respectively.

^d F^* is the clay-corrected formation factor.

Since Eq. (4.37) is determined from sandstone core plugs fully saturated with water, the presence of oil will affect the value of $T_{2,LM}$, and hence the permeability prediction. In this situation, other permeability predictors, as discussed below, using irreducible water saturation S_{wirr} as input are frequently used.

Coates/Timur equation

Multiple studies have shown that the permeability in clastic reservoirs is related to the irreducible brine saturation (S_{wirr}). Related quantities that can be estimated from NMR are the Bulk Volume Irreducible water BVI (or ϕS_{wirr}), and the Free Fluid Index (FFI) (or $\phi(1 - S_{wirr})$).

Timur [214] developed a permeability estimator based on numerous core measurements. It has the following form:

$$k = 10^4 \frac{\phi^{4.5}}{S_{wirr}^2} \quad (4.39)$$

where ϕ and S_{wirr} are in fractions, and k is in mD (milli-Darcy).

Coates [63] developed a widely used equation based on the irreducible brine saturation. There are several versions of his equation. The most commonly used one is:

$$k = ((\phi/10)^2 (FFI/BVI))^2 \quad (4.40)$$

where ϕ , FFI, and BVI, are in p.u. (porosity unit), and k is in mD.

This kind of empirical form for permeability prediction can be traced back to the Kozeny equation [53], i.e.,

$$k = A \frac{\phi^3}{S^2}, \quad (4.41)$$

where ϕ is porosity, S is the surface area per unit bulk volume, and A is an empirical constant. The surface area S in the denominator is related to the irreducible water saturation or BVI.

Summation of echoes

The summation of echo amplitudes is mathematically related to the mean relaxation time, $\langle T_2 \rangle$. Sezginer et al. [199] showed that

$$p = \sum_{i=1}^N g_i \approx \sum_{i=1}^N \epsilon_i + \text{HI} \phi \langle T_2 \rangle / t_E \quad (4.42)$$

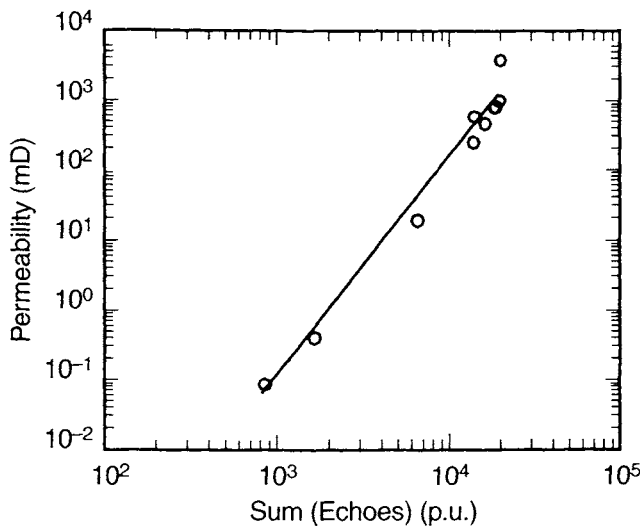


Figure 4.20: Measured permeability versus sum of echoes for 10 quarried sandstone samples. $TE=0.2$ ms. (From Sezginer et al. [199].)

where p is the summation of echoes; g_i , the amplitude of the i -th echo; ϵ_i , the noise level of i -th echo; HI , the hydrogen index; ϕ , the porosity; $\langle T_2 \rangle$, the mean relaxation time; t_E , the echo spacing, and N the number of echoes.

$\langle T_2 \rangle$ is somewhat different from the commonly used logarithmic mean, $T_{2,LM}$. It is related to the surface to volume ratio (with strong weight for large pores) of the rock formation; thus, it is related to permeability. From laboratory experiments on quarried sandstones, Sezginer et al. [199] found a good correlation (Figure 4.20), and suggested the following relations for $TE=0.2$ ms:

$$k \approx 6.18 \times 10^{-12} p^{3.52} \text{ for } N = 600 \quad (4.43)$$

$$k \approx 4.58 \times 10^{-12} p^{3.13} \text{ for } N = 1200 \quad (4.44)$$

where k is the permeability in mD, and p is in p.u.

Permeability prediction for carbonates

Carbonates not only occur in a wide variety of pore sizes and shapes, but also exhibit a heterogeneous distribution of pore types at all core and log-

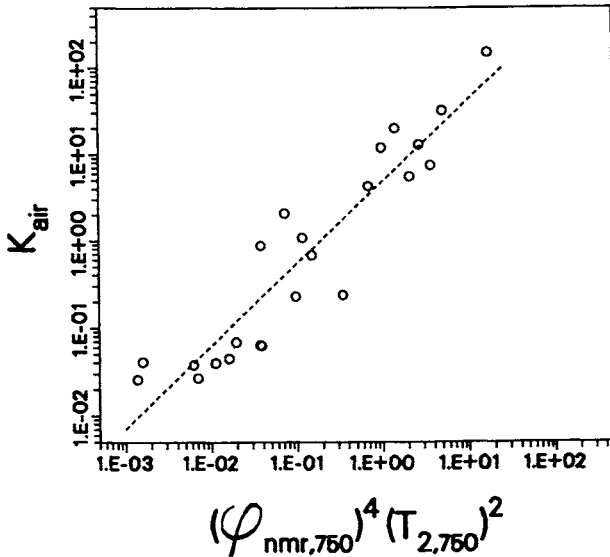


Figure 4.21: Crossplot of air permeability K_{air} versus $(\phi_{NMR,<750})^4 (T_{2,<750})^2$ for a suite of carbonates where the porosity and T_2 above 750 ms are not included. (From Chang et al. [56].)

ging measurement scales. Chang et al. [56] has published studies of NMR T_2 measurements on samples 100% brine saturated and desaturated by centrifugation to approximately S_{wirr} . Based on these studies, they infer that all signal above 750 ms is due to vugs, and that if the vuggy porosity is not included for ϕ and T_2 , then the permeability equation $k = \phi^4 T_{2,LM}^2$ works quite well. A crossplot of air permeability K_{air} versus $(\phi_{NMR,<750})^4 (T_{2,<750})^2$, for a suite of carbonates from their work, is shown in Figure 4.21. Their measurements were made in homogeneous magnetic field. The NMR logging tools usually have significant field gradients which will reduce the 750 ms cutoff substantially when the vugs are brine filled and to a lesser extent if the vugs are filled with light crude.

4.3.4 Wettability

Wettability is defined as “the tendency of one fluid to spread on or adhere to a solid surface in the presence of other immiscible fluids” [69]. The spreading on or adherence to a solid surface is commonly described as wetting. It can

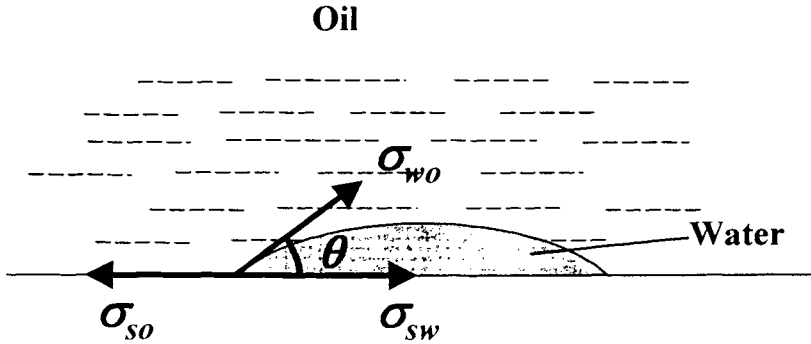


Figure 4.22: Schematic diagram of a water-oil-solid interface showing the equilibrium of interfacial forces, where σ_{so} , σ_{sw} , and σ_{wo} are the interfacial tensions between the solid and oil, solid and water, and water and oil, respectively.

be measured by *adhesion tension*, a function of the interfacial tension which determines which fluid will preferentially wet the solid.

As shown in Figure 4.22, two immiscible liquids, oil and water, are in contact with a solid. The contact angle θ is measured through the denser liquid phase and ranges from 0° to 180° . The adhesion tension A_T is defined as

$$A_T = \sigma_{so} - \sigma_{sw} = \sigma_{wo} \cos \theta_{wo} \quad (4.45)$$

where σ_{so} is the interfacial tension between the solid and lighter fluid, oil; σ_{sw} is the interfacial tension between the solid and denser fluid, water; and σ_{wo} is the interfacial tension between fluids, water and oil.

A positive value of adhesion tension indicates that the denser liquid preferentially wets the solid surface, whereas a negative value indicates the lighter liquid preferentially wets the surface. A zero value indicates both liquids have an equal affinity for the surface. A reservoir rock can be water-wet, or oil-wet, or having both conditions coexist, i.e., a mixed wettability situation. Wettability of reservoir rocks can have dramatic effects on the effective per-

meability of oil⁵ and the efficiency of oil recovery.

There is abundant literature on wettability studies [9, 190, 170, 130]. There are different methods for measuring wettability [9], some of them quite elaborate and time consuming. Although these methods show general agreement in discriminating oil-wet versus water-wet media, there is no quantitative correspondence between these measures.

NMR relaxation is a convenient method for characterizing wettability of a pore fluid phase. When a non-wetting phase becomes partially-wetting, its relaxation time typically becomes shorter as compared to its value for the bulk phase due to surface relaxation. There have been numerous studies on the wettability problem using NMR methods [37, 191, 224, 107, 74, 29, 105, 106, 174, 27, 155, 204, 232]. There is, however, no generally agreed upon method for quantifying the wettability. These studies are largely confined to laboratory core measurements. Applications of NMR logs for wettability analysis have been scanty, and require careful coordination of core measurements and log interpretation.

Brown and Fatt [37] studied the wettability problem of reservoir rocks using an NMR method with mixtures of preferentially oil wet and preferentially water wet sands. Their result for relaxation rates of fully water saturated sand mixtures containing various percentages of preferentially oil wet sands is shown in Figure 4.23. The linear relationship clearly suggests that the water wet surface has a higher surface relaxivity than the oil wet surface, and the linear variation of the relaxation rate simply reflects the abundance of the oil wet surfaces. To apply such a finding to core analysis, they suggested using a procedure of measuring the relaxation rates of core at an unaltered wettability state, 100 percent water-wet state, and 100 percent oil-wet state. Then the percentage of the oil-wet surface for the initial mixed wettability state is determined from the straight line drawn between the 100 percent water-wet and 100 percent oil-wet states.

However, the application of such reasoning to the amount of wettability from NMR relaxation rate measurements is complicated by the pore size effect. Except when there is only one pore size, the relaxation rate measured at $t=0$ is an average relaxation rate of all pore sizes. If the oil-wetting occurs preferentially for larger pores, the variation of the relaxation rate may not be linear as that shown in Figure 4.23. Then the interpretation of the fractional

⁵The effective permeability is a relative measure of the conductance of the porous medium for one fluid phase when the medium is saturated with more than one fluid.

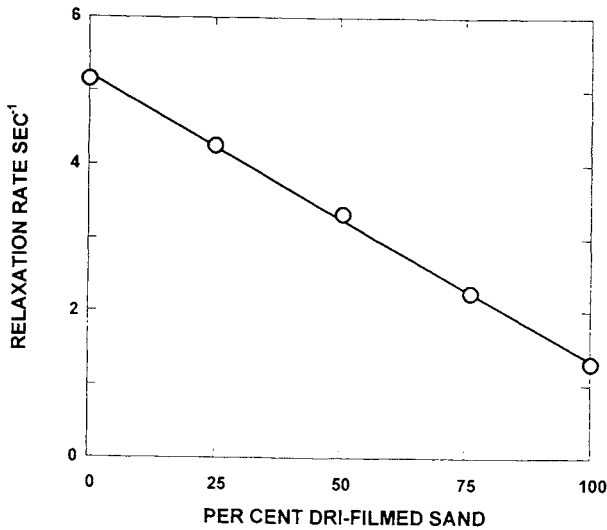


Figure 4.23: Relaxation rates for sand mixtures containing various percentages of oil wet sand. (From Brown and Fatt [37].)

wettability of a core becomes complicated and can be ambiguous.

Borgia et al. [29] and Howard [105, 106] chose chalk samples for the wettability study because of the narrow distribution of pore sizes for chalk samples making the interpretation of experimental results easier. Amott index [7] (the ratio of the spontaneous imbibition of water to the amount of imbibition plus forced displacement by centrifuge) was used in their analysis either to characterize the wettability of the rock samples or to correlate to an NMR-based wettability index. Howard suggested an NMR based wettability index as follows:

$$T_n = CS_w^b \quad (4.46)$$

where T_n is the normalized relaxation time of the water-filled component compared to a fully saturated state (i.e., $T_n = T_{2,s_w}/T_{2,s_w=1}$ where T_{2,s_w} and $T_{2,s_w=1}$ are the T_2 peak of water at partially and fully water-saturated states, respectively), C is an empirical constant, and the exponent b is correlated with wettability. For a fully water wet core, b is equal to 1. A decrease in b indicates weaker proton interaction with the pore wall, i.e., less water wet, at a given water saturation.

Zhang et al.[232] studied the wettability of some sandstones with an NMR method and found that some of them became mixed-wet after being saturated with refined or crude oil and aging for a certain length of time. Some of the pore-lining chlorite flakes apparently became oil wet and shielded the water from having contact with the pore walls.

Wettability is an important parameter in oil recovery. NMR offers a convenient technique for characterizing wettability. However, a quantitative and generally accepted method for measuring wettability has yet to be developed.

4.4 MAS NMR

The Magic Angle Spinning (MAS) technique is used routinely in solid state NMR to average out chemical shift anisotropy and internuclear dipolar interactions (see, for example, Mehring [160]). However, the application of such a technique to the study of fluids contained in rocks is relatively recent (Xiao et al. [228], de Swiet et al. [71], Wilson and LaTorraca [225]). There are many areas yet to be explored, such as oil viscosity, preferential wettability, etc., which may develop into a powerful technique for petrophysical analysis.

In fluid-saturated rocks, the paramagnetic impurities residing at the pore wall, as well as the magnetic susceptibility contrast between the solid matrix and the pore fluid, create a rather heterogeneous local field environment. Such field inhomogeneity leads to a broad featureless line in ^1H NMR spectra and renders it useless in identifying or quantifying the oil and water contents in the rocks. The local field $\mathbf{B}_{\text{loc}}$, in a dipole approximation, is given by

$$\mathbf{B}_{\text{loc}}(\mathbf{r}) \cdot \hat{z} = \frac{B_0}{4\pi} \int \chi(\mathbf{r}') \frac{(3 \cos^2 \theta - 1)}{|\mathbf{r} - \mathbf{r}'|^3} d\mathbf{r}' \quad (4.47)$$

where $\chi(\mathbf{r}')$ is the local magnetic susceptibility in the rock, B_0 the external applied field in the z direction, and θ is the angle that $\mathbf{r} - \mathbf{r}'$ makes with the applied field as shown in Figure 4.24.

When a fluid-saturated rock is placed in a rotor inclined at an angle θ with respect to the applied field B_0 in the z direction, the proton spin inside the molecules is aligned in the z direction. As the sample starts to rotate about the rotor axis, the proton spins remain aligned in the z -direction, whereas the surrounding paramagnetic atoms residing either on the pore wall or solid matrix start to rotate. Thus the proton spins experience a time varying local field which is rotating about the rotor axis. The NMR signals that

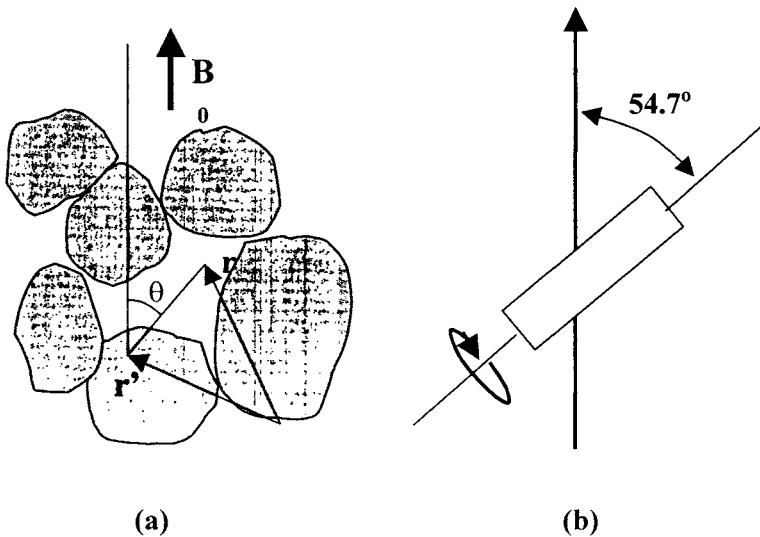


Figure 4.24: The local dipolar field in a rock (a), and the magic angle spinning (b) where the rock sample is placed inside a rotor spinning about its axis which makes an angle of 54.7° with respect to the applied field B_0 .

we measure will be a time average of these varying fields. If the magnetic susceptibility $\chi(r')$ is isotropic, all of these will average to zero, except the contributions of paramagnetic atoms lying along the rotor axis. But those contributions are also made to vanish, as we purposely set the inclined angle $\theta = 54.7^\circ = \cos^{-1}(1/\sqrt{3})$ which nullifies the dipolar field given by Eq. (4.47). Hence, this technique is called magic angle spinning (MAS).

As shown in the left side of Figure 4.25, when the rotor is stationary, the ^1H NMR spectrum of a water-saturated Berea sandstone shows a broad featureless peak. When the rotor spins at 1 and 2.5 kHz, the broad spectrum narrows to a sharp peak with many spinning sidebands, spaced at the rotor frequency. The existence of the spinning sidebands can be understood as follows: Recall that the spectrum is the Fourier transform of the FID after the application of a 90° rf pulse. As the spins are precessing in the xy plane and signals are being collected, those with the Larmor frequency determined by B_0 will be stationary in a coordinate frame rotating about the z -axis with that frequency. All spins with different local Larmor frequencies will be rotation-averaged except those whose local Larmor frequency is higher or lower than

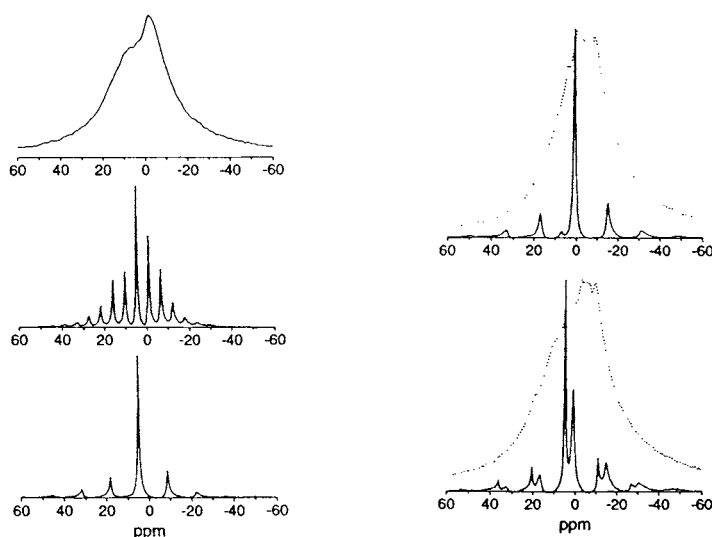


Figure 4.25: The effect of MAS on ^1H NMR spectra of water-saturated Berea sandstone is shown on the left. From top to bottom are spectra of spinning the sample at 0 Hz, 1 kHz, and 2.5 kHz, respectively. On the right, it shows the ^1H NMR spectra of Berea sandstone saturated with a light crude oil (top), and a mixture of crude and water (bottom). The rotor spinning speed was 2 kHz in both cases. The dotted line shows the spectra in stationary state, and the solid line shows the spectra at MAS condition. (From de Swiet et al. [71].)

the main Larmor frequency by an exact multiple of the rotor frequency. They will also be stationary in the rotating frame indistinguishable from those with the main Larmor frequency and be collected in the FID. Thus, they will also show up in the Fourier transform of the NMR signal. As the rotor spinning frequency increases, the spinning sidebands move out from the main peak and eventually disappear.

A more interesting aspect of the MAS technique is its application to the wettability issue, as demonstrated by the work of de Swiet et al. [71] shown in the right of Figure 4.25. The top of the figure shows the NMR spectrum of a Berea sandstone, saturated with a light crude of 12 centipoise viscosity, both in stationary state (dotted line) and at a spinning speed of 2 kHz (solid line). The spectrum of the oil shows a large peak in the aliphatic region

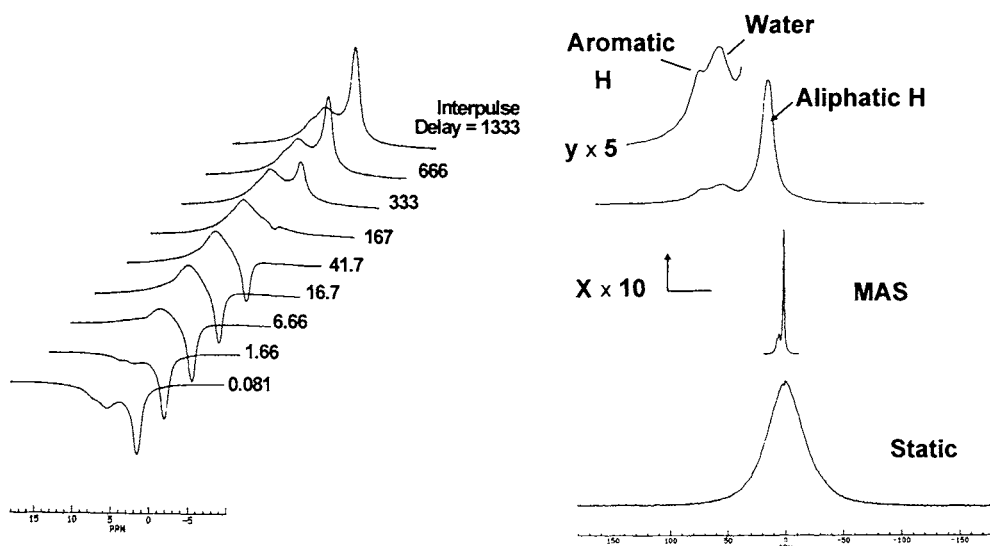


Figure 4.26: The ^1H NMR spectra of a T_1 inversion recovery sequence at different τ (ms) in the MAS experiment for a diatomite sample. As shown at the right of the figure, from bottom to top, the ^1H NMR spectra of the sample when it is stationary, when it is under the MAS condition, and the enlarged plots ($\times 10$ and then $\times 5$) of the observed peak in the MAS experiment. The major peak is from the aliphatic H, and to the left of the major peak are the water and aromatic H at much smaller amplitudes. (From Wilson and LaTorraca [225].)

and a small peak in the aromatic region (i.e., the tiny bump at the left tail of the central peak), as expected for a light crude. This information is obviously unavailable in the broad peak exhibited by the stationary sample. The bottom of the figure shows the spectrum of a Berea sandstone saturated with a mixture of crude and water. The spectrum exhibits an aliphatic peak and a much stronger water peak which obscures the very small aromatic peak near its base. Assuming the ^1H density is the same in oil and water, de Swiet et al. calculated the oil:water ratio within the sample to be $0.95 : 1.0 \pm 0.1$. Considering the sample was prepared by immersion in a mixture of oil:water ratio of 3:1, they concluded the sample was preferentially water wet.

In fact, the wettability issue can be more quantitatively studied by measuring the T_1 relaxation of the sample under the spinning condition. Wilson

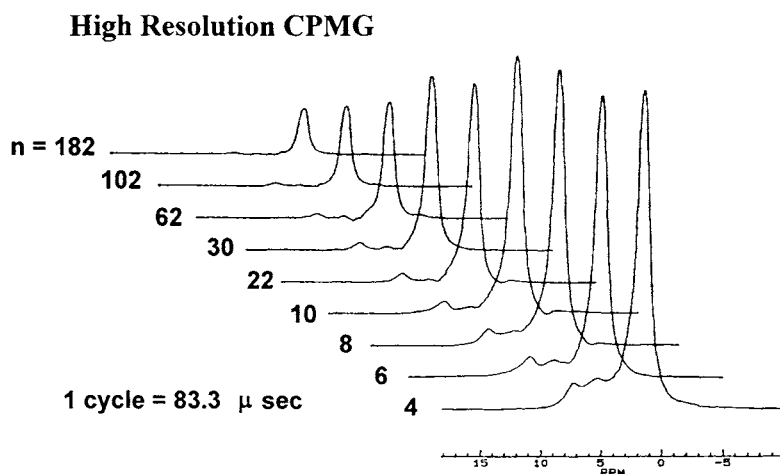


Figure 4.27: The ^1H NMR spectra of a T_2 CPMG sequence at selected n -th spin echo in the MAS experiment for a diatomite sample. The major peak is from the aliphatic H, and the minor peaks of water and aromatic H are clearly visible at the left shoulder of the major peak. (From Wilson and LaTorraca [225].)

and LaTorraca [225] show measurements of T_1 (Figure 4.26) and T_2 (Figure 4.27) for diatomite samples under the MAS condition. The high resolution T_1 method is the common inversion recovery sequence with a Fourier transform to yield the spectra. The CPMG sequence was obtained with π pulses synchronized with MAS spinning, i.e., the echo spacing being the same as the spinning cycle, with a Fourier transform to yield the spectrum at selected n -th spin echoes. As shown in Figure 4.27, the major peak is from the aliphatic H, and to the left of the major peak are the water and aromatic H at much smaller amplitudes. The amplitude of each peak can be easily tracked, thus its relaxation behavior can be analyzed independently.

MAS NMR has been shown so far to be a promising technique for core analysis. The mechanical effect of MAS was qualitatively investigated by Xiao et al. [228]. The centrifugal forces may overcome capillary pressure and cause pore fluid to slosh around creating undesirable artefacts for the experiments. Further work is needed so that proper sample preparation

procedures can be developed to minimize such an effect.

4.5 NMR core imaging

NMR imaging is mainly used for core imaging in a similar way as the X-ray CAT (Computer-Aided Tomography) scans are used in core studies. The purpose is to understand the fluid distributions within a core before, during, and after a core-flood experiment. The X-ray CAT scan image is based on the contrast in X-ray absorption in different parts of the core. The images depend largely on the high X-ray absorption of the minerals in the solid matrix, not that of the pore fluids. In contrast, all the signal in NMR images comes from the hydrogen nuclei in water or oil in the pore space. The matrix does not contribute to the signal at all. The two techniques are thus complementary.

Unlike the petrophysical applications of other NMR properties, which have a close relationship to rock properties such as irreducible water saturation, pore size distribution, and permeability, NMR core imaging is at present mostly limited to studies of fluid distributions within the pore space. In the future, NMR core imaging may very well develop into a tool for wettability studies and other applications. Because of its importance in core study, as well as in biology and medicine, we shall briefly outline its principles here. For a detailed account, the readers should refer to Callaghan [49].

NMR imaging can perhaps be best understood by first considering an example of one-dimensional imaging: In a T_2 measurement, where the applied field H_0 lies along z and the Larmor angular frequency is $\omega_0 = \gamma H_0$, the magnetization vector is first tipped from the z axis into the xy plane. We then turn on a constant field gradient G_z in the z direction (i.e., the gradient is in the same direction as the spin polarizing field itself) which is so large that it overwhelms the small field inhomogeneities of the instrument and forces different spins to precess differently according to their z coordinate: Each spin precesses at some frequency ω , which differs from ω_0 by

$$\Delta\omega \equiv \omega - \omega_0 = \gamma G_z z. \quad (4.48)$$

Note that this offset can be either positive or negative, since each of the contributing factors G_z , z can be either positive or negative (the gyromagnetic ratio γ for proton is, of course, positive). If the constant field gradient G_z is applied for a time t , those spins will acquire an extra phase $\gamma G_z z t$. Thus,

the free induction decay (FID) signal obtained from the sample, while the gradient G_z is on, is given by

$$S(t) = \int \rho(z) \exp(i\gamma G_z z t) dz. \quad (4.49)$$

The inverse Fourier transform of $S(t)$ is then proportional to the one-dimensional proton density profile $\rho(z)$ along the z axis:

$$\rho(z) \sim \int S(t) \exp(-i\gamma G_z z t) dt. \quad (4.50)$$

Extending this observation to three dimensions, when the pulse field gradient $\mathbf{G}$ is a three dimensional vector which is turned on for a time t and can point in *any direction*, the spins at $\mathbf{r}$ now precess at an excess frequency of

$$\Delta\omega = \omega - \omega_0 = \gamma \mathbf{G} \cdot \mathbf{r}. \quad (4.51)$$

Thus, the signal from all $\rho(\mathbf{r})$ is given by

$$S(t) = \int \int \int \rho(\mathbf{r}) \exp(i\gamma \mathbf{G} \cdot \mathbf{r} t) d\mathbf{r}. \quad (4.52)$$

From Eq. (4.52), we can define a “reciprocal space vector” $\mathbf{k}$ in three dimensions as

$$\mathbf{k} = \gamma \mathbf{G} t. \quad (4.53)$$

This allows us to introduce the conjugate pair of functions, $S(\mathbf{k})$ and $\rho(\mathbf{r})$, where $S(\mathbf{k})$ is the spatial Fourier transform of the local proton density in the system $\rho(\mathbf{r})$. These two quantities are related by means of Fourier transformations:

$$S(\mathbf{k}) = \int \int \int \rho(\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}, \quad (4.54)$$

$$\rho(\mathbf{r}) = \int \int \int S(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{r}} d\mathbf{k}. \quad (4.55)$$

This concept of “ $\mathbf{k}$ ” space was first suggested by Mansfield [150, 151, 152, 153]. With this definition, $\mathbf{k}$ can be made to span the entire three-dimensional space by varying $\mathbf{G}$ and t . Thus, we have constructed a scheme for measuring the “signal” $S(\mathbf{k})$ as a function of $\mathbf{k}$. Using the inverse Fourier transformation of Eq. (4.55), $S(\mathbf{k})$ can be processed to produce the NMR image of $\rho(\mathbf{r})$.

In the above discussion of Eq. (4.52), we neglected the T_2 spin relaxation. When this is not justified, special pulsing schemes should be employed in order to obtain a faithful NMR image.

To illustrate how this is typically done in practice, we consider the example outlined in Figure 4.28: First, a 90° rf pulse, with a sinc function ($\sin x/x$) envelope and a mean carrier frequency ω_0 , generates a square shaped slice in the frequency domain *while a large field gradient G_z is applied* along z , in the same direction as the mean spin polarizing field $H_0\mathbf{e}_z$. This procedure selects a thin slab or slice of the sample, parallel to the xy plane. The spins in this slice have a narrow range of Larmor precession frequencies centered around ω_0 and are tilted by the rf pulse into the xy plane. The thickness of this slice is determined by the length of the sinc function and the magnitude of G_z . Only the spins in this thin slice are in resonance with the rf pulse and are, therefore, tipped to the xy plane. Some phase spreading of these spins will occur as a result of the large gradient G_z . The field gradient is then turned off, after which a 180° spin inverting rf pulse is applied. Then the *same field gradient G_z* is turned on again, but for a different length of time, in order to refocus the phases of the precessing spins in the selected slice in an optimal fashion. The purpose of the 180° rf pulse and the subsequent pulse field gradient G_z is to make all the precessing spins in the selected slice have nearly the same phase and frequency, in preparation for the subsequent phase and frequency encoding.

At this juncture, a pulse *field gradient G_y* is turned on, along y , for a time t_y . This causes each spin to precess at a rate that depends on its y coordinate. Thus, it acquires a phase which depends on that coordinate. This corresponds to shifting the origin in $\mathbf{k}$ space by an amount $k_y = \gamma G_y t_y$ along y . This is commonly referred to as “phase encoding”.

After that, the so-called “*read*” gradient G_x is turned on, along x , and the FID signal is collected. The gradient G_x now causes the spins with the same y coordinate, which used to have the same rate of precession, to be split into groups with different Larmor frequencies, varying in linear fashion with x . This part of the procedure is called “frequency encoding”.

While this gradient G_x is left on, the FID signal is collected from the entire system, as a function of the time t_x elapsed since that gradient was turned on. The signal comes from *only those spins* in the original xy slice that were tipped to the xy plane in the first place, all of which have the *same value of z !* Each digitized signal of the FID corresponds to a particular value of t_x , and thus to a particular value of $k_x = \gamma G_x t_x$. Therefore the signal,

recorded as function of t_x , represents a trace along the k_x direction for all those k_x values and with a fixed value of k_y . The digitized FID is therefore a sequence of values of $S(k_x, k_y, z)$, with k_x ranging from 0 to $k_{x,max}$ and at fixed values of k_y and z . By repeating this measurement, using different values of G_y and a fixed time interval t_y , we can collect $S(k_x, k_y, z)$ for *all values of k_x and k_y at one fixed value of z* . A 2D NMR image can then be obtained by using the 2D version of the inverse Fourier transformation of Eq. (4.55). In order to obtain the full three-dimensional image, the mean carrier frequency of the initial 90° soft (sinc function) rf pulse is varied to select different xy slices in the sample.

Numerous other field gradient and rf pulsing schemes or sequences can be used for obtaining NMR images of $\rho(\mathbf{r})$. The basic idea is always to find an efficient way of measuring $S(k_x, k_y, z)$ at k_x and k_y values which span that entire 2D subspace of $\mathbf{k}$ space and at a fixed value of z . Figure 4.29 shows low resolution NMR images of the fluids in thin slices through rock samples.

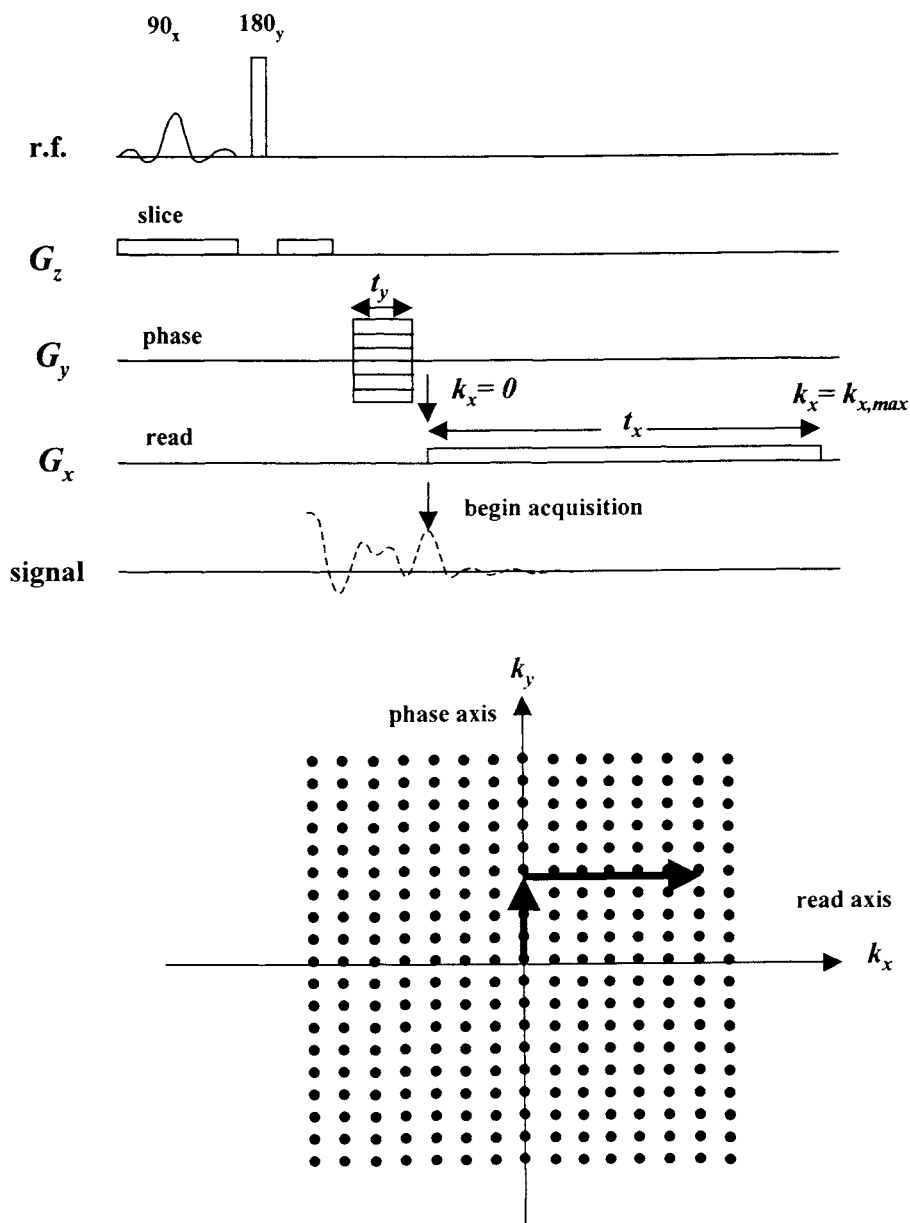


Figure 4.28: The pulsing sequence of getting a 2D NMR image. The bottom of the figure illustrates the path of scanning the $\mathbf{k}$ vector in $\mathbf{k}$ space. (From Callaghan [49]. Redrawn from Figure 3.14 of Ref. [49] with author's permission.)

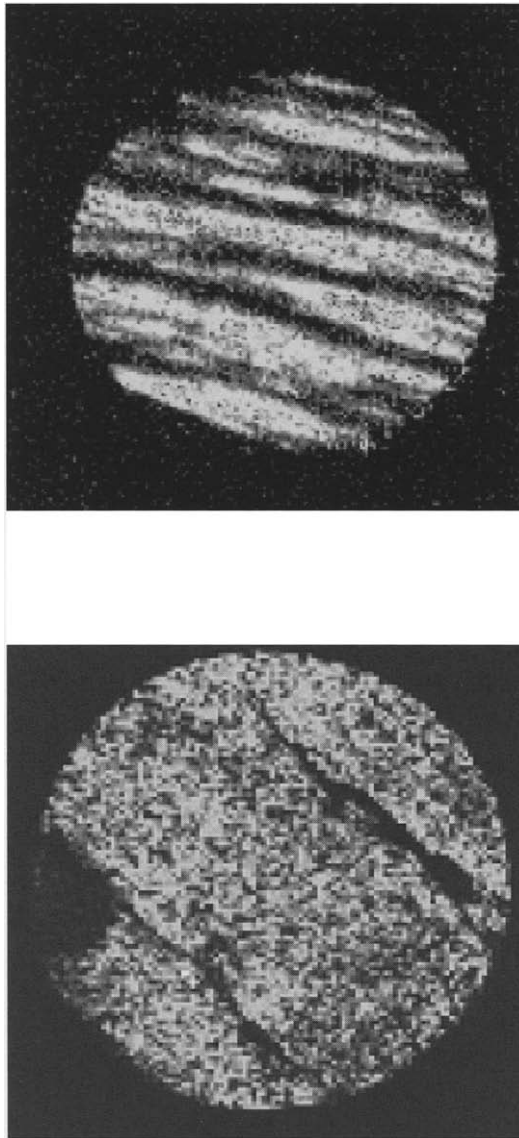


Figure 4.29: NMR images of thin slices through rock samples. The light color represents higher porosity.